Energy 242 (2022) 123026

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Strategic bidding of an energy storage agent in a joint energy and reserve market under stochastic generation

Christos N. Dimitriadis, Evangelos G. Tsimopoulos, Michael C. Georgiadis*

Aristotle University of Thessaloniki, Department of Chemical Engineering, School of Engineering, P.O. Box 455, Thessaloniki, 54124, Greece

ARTICLE INFO

Article history:
Received 11 June 2021
Received in revised form
3 December 2021
Accepted 23 December 2021
Available online 28 December 2021

Keywords:
Mathematical problem with equilibrium constraints
Energy storage
Joint electricity-reserve market clearing
Reserve activation
Network congestions
Wind power generation uncertainty

ABSTRACT

This work presents a bi-level optimization model for a price-maker energy storage agent, to determine the optimal hourly offering/bidding strategies in pool-based markets, under wind power generation uncertainty. The upper-level problem aims at maximizing storage agent's expected profits, whereas at the lower-level problem, a two-stage sequential market clearing procedure takes place. The first stage refers to a jointly cleared energy and reserve day-ahead market, deriving the optimal quantities for energy dispatch and reserve procurement, while at the second stage, a real-time energy-only market settlement is realized, considering plausible wind power generation scenarios. Real-time decisions include activated upward and downward reserves, already procured in the day-ahead market. The bi-level model is initially transformed into a mathematical program with equilibrium constraints -by recasting the lower-level problem using its Karush-Kuhn-Tucker optimality conditions-, which is further reduced into a mixed integer linear program. The proposed mathematical framework is applied to a 6-bus power grid, incurred by network transmission constraints. Numerical simulations illustrate an explicit perspective of the potential arbitrage opportunities for the storage agent, when acting strategically in energy and reserve markets, under wind generation increment scenarios and plausible transmission line congestions.

© 2021 Elsevier Ltd. All rights reserved.

1. Introduction

The ever-increasing necessity for assertive penetration of non-dispatchable energy sources, such as wind, due to their beneficial financial and environmental impact, has contributed to the critical deployment of the role, Energy Storage Systems (ESS) own in electricity markets' contemporary scheme. An increased contribution of renewable energy sources' (RES) share into the electricity grid's mixture, can provide a decisive prospect for suppressing wholesale energy prices, through outplacing high-cost generators from the merit-order. Furthermore, this RES contribution can lead to downscaling dependence on dispatchable generation (fossil fuels) and confining greenhouse gas emissions [1]. Their erratic and fluctuating nature regarding their generation levels though, can originate concerns about the reliability of the power grid comprised in great measure of these sources [2].

The significant progress that has been achieved in energy storage technologies and their applications can address the aforementioned issues, leading to a rapid decarbonization, while providing ancillary services such as reserves, to guarantee the stability of supply and demand equilibrium in power systems [3]. Apart from the implicitly advantageous contribution to the electrical grid, nowadays a large-scale privately owned ESS can enable significant economic opportunities for an investor who seeks to maximize his expected profit, participating in a liberalized energy market environment [4]. In particular, storage systems can act strategically and capitalize their (dis)charging decisions by arbitraging; procuring and storing electricity when off-peak electricity prices occur and trade it back in the market when electricity prices rise [5]. A case study conducted by McConnell, Forcey and Sandiford [6] demonstrated that the above ESS's operation can stimulate a significant competitive advantage over other involved generators in Australian National Electricity Market (NEM), an especially well designed and regulated energy-only market, that is characterized by high wind power generation insertion [7].

In today's energy markets, each strategic agent's practices are typically modelled through a bi-level optimization problem approach [8], whose upper-level objective is concentrated on agent's profit seeking behavior, whereas lower-level problem

* Corresponding author.
E-mail address: mgeorg@auth.gr (M.C. Georgiadis).

https://doi.org/10.1016/j.energy.2021.123026
0360-5442/© 2021 Elsevier Ltd. All rights reserved.

Nomenclature

Indices and sets

i	Index of conventional generating units
j	Index of wind generating units
d	Index of demands
s	Index of storage systems
t, k	Indices of time periods
n, m	Indices for system buses
ω	Index of scenarios
$I\alpha N$	Set of indices of conventional generating units located at bus n
JaN	Set of indices of wind generating units located at bus n
DaN	Set of indices of demands located at bus n
SaN	Set of indices of storage systems located at bus n
NaM	Set of buses n connected with bus m

Superscripts

ch	Charging mode of storage system s
dis	Discharging mode of storage system s
$\uparrow$	Upward reserve
$\downarrow$	Downward reserve

Parameters

c_s	Marginal cost of storage system s at (dis)charging mode (€/MWh)
G_s^{max}	Maximum (dis)charging capacity of storage system s (MW)
RC_s^{max}	Maximum upward/downward reserve capacity of storage system s at (dis)charging mode (MW)
SOC_s^{max}	Maximum storage capacity of storage system s (MW)
SOC_s^{ini}	Initial reserved energy of storage system s (MW)
η	(Dis)charging efficiency of storage system s
$B_{n,m}$	Susceptance of line n-m
R_t	Upward/downward reserve requirements of the market at time interval t (MW)
P_i^{max}	Maximum capacity at conventional generating unit i (MW)
$L_{d,t}^{max}$	Maximum load of demand d at time t (MW)
RCc_i^{max}	Maximum upward/downward reserve capacity of conventional generating unit i (MW)
$RCd_{d,t}^{max}$	Maximum upward/downward reserve capacity load of demand d (MW) at time t
$W_{j,t}^{max}$	Available generation of wind unit j at time t (MW)
$T_{n,m}^{max}$	Transmission capacity of circuit line n-m
π_ω	Occurrence probability of scenarios ω
$VOLL_{d,t}$	Value of lost load d at time t (€/MWh)
$W_{j,t,\omega}^{RT}$	Scenario dependent generation of wind unit j at time t (MWh)
$c_{i,t}$	Cost offer for dispatched energy of conventional unit i at time t (€/MWh)
$c_{i,t}^{res}$	Cost offer for procured reserves of conventional unit i at time t (€/MWh)
$c_{j,t}^{w}$	Cost offer for dispatched energy of wind generator j at time t (€/MWh)
$u_{d,t}$	Utility cost for consumed energy of demand d at time t (€/MWh)
$u_{d,t}^{res}$	Utility cost for procured reserves of demand d at time t (€/MWh)

Upper-level variables

$soc_{s,t}$	Energy stored for storage system s at time t (MWh)
$x_{s,t}$	Binary decision variable indicating (dis)charging mode of storage system s at time t
$\overline{G}_{s,t}$	Energy amount bid by storage system s at time t in (dis)charging mode (MW)
$\overline{rc}_{s,t}$	Reserve capacity bid by storage system s at time t in (dis)charging mode (MW)
$O_{s,t}$	Price bid by storage system s at time t in (dis)charging mode (€/MWh)

Lower-level variables

$P_{i,t}^{DA}$	Scheduled production of conventional generating unit i at time t in DA market (MWh)
$rp_{s,t}$	Upward/downward reserve procurement from storage unit s in time t in (dis)charging mode in DA market (MW)
$L_{d,t}^{DA}$	Energy consumed by load d at time t in DA market (MWh)
$rpd_{d,t}$	Reserve procurement from demand d at time t in DA market (MW)
$G_{s,t}^{DA}$	Scheduled energy consumption of storage system s at time t in (dis)charging mode in DA market (MWh)
$rpc_{i,t}$	Reserve procurement from conventional generating unit i at time t in DA market (MW)
$W_{j,t}^{DA}$	Scheduled energy production of wind generating unit j at time t in DA market (MWh)
$\lambda_{n,t}^{DA}$	Clearing price in DA market at bus n at time t (€/MWh)
$\delta_{n,t}^{\circ}$	Voltage angle at bus n in DA market
$\lambda_t^{\uparrow}$	Clearing price for upward reserve procurement at time t (€/MWh)
$\lambda_t^{\downarrow}$	Clearing price for downward reserve procurement at time t (€/MWh)
$rac_{i,t,\omega}$	Reserve activation from conventional generating unit i at time t under scenario ω (MWh)
$rad_{d,t,\omega}$	Reserve activation from demand d at time t under scenario ω (MWh)
$L_{d,t,\omega}^{sh}$	Load shedding of demand d at time t under scenario ω (MWh)
$ra_{s,t,\omega}$	Reserve activation from storage system s at time t under scenario ω in (dis)charging mode (MWh)
$\lambda_{n,t,\omega}^{RT}$	Clearing price in RT market at bus n at time t under scenario ω (€/MWh)
$\delta_{n,t,\omega}$	Voltage angle at bus n at time t under scenario ω
$W_{j,t,\omega}^{sp}$	Power spillage from wind generating unit j at time t under scenario ω (MWh)

Dual Variables

α, θ	Dual variables corresponding to conventional generating unit i in DA and RT market
β, μ	Dual variables corresponding to demand d in DA and RT market
γ, ν	Dual variables corresponding to storage system s in DA and RT market
ε, ξ	Dual variables corresponding to wind generating unit j in DA and RT market
ζ, φ	Dual variables corresponding to transmission lines of buses n, m in DA and RT market

describes the market clearing procedure, conducted by the independent system operator (ISO) [9]. Ruiz and Conejo [10] adopted these modelling tendencies and introduced a bi-level optimization problem, applied in a transmission constrained network, to derive the optimal hourly offering strategies of a price-maker power generator, in a pool-based market.

Baringo and Conejo [11] contemplated a transmission constrained power grid formulating a stochastic bi-level model. The upper-level objective aims at maximizing a strategic wind power investor's expected profit, while the lower-level presents the market clearing process under variable load and wind generation scenarios. Morales et al. [12] proposed a single market settlement clearing algorithm for a transmission constrained power system with inelastic loads, incorporating a significant number of wind producers. Nasiri et al. [13] introduced a bi-level mathematical approach to model the strategic scheduling of a price-maker multi-energy system (MES) in a day-ahead pool-based market. A hybrid modelling framework that combines robust optimization and stochastic programming was implemented to deal with the uncertain nature of wind power. Recently, Tsimopoulos and Georgiadis [14,15] considered a bi-level multi-block optimization problem for providing the optimal hourly offers for a conventional power producer, under variable wind power generation scenarios.

Most state-of-the-art approaches to modelling the integration of electricity storage systems into the modern energy-only markets, include the development of mathematical program with equilibrium constraints (MPEC) models and their consequent reformulation to mixed integer linear programming (MILP) problems. The proposed mixed complementarity problem by Awad et al. [16] integrates a large-scale ESS into the market's energy mixture, ensuring higher levels of stability and efficiency for the grid and addresses the impact of ESS's location and size on market's equilibrium. Pandžić and Kuzle [17] presented a bi-level program, applied to the 24-bus IEEE system, considering the impact of an energy storage system on conventional generators' performance in electricity markets. Ye, Papadaskalopoulos and Strbac [18] introduced a bi-level problem to provide quantitative evidence of the energy storage's positive contribution in limiting strategic producers' power, in the day-ahead market.

Wang et al. [19] modelled the optimal energy storage operator's bidding strategy, recommending a look-ahead technique, which addresses both day-ahead and following day market by adjusting the ESS's state-of-charge at the end of the former day. Gomes et al. [20] proposed a two-stage stochastic MILP, for deriving optimal bidding strategies for a power generator owing a wind and a photovoltaic system with energy storage and participating in the Iberian day-ahead energy market. Ding, Pinson and Hu [21] developed a stochastic programming problem for generating offering strategies for a wind farm-energy storage system in day-ahead and real-time market, under wind generation uncertainty.

Zou et al. [22] formulated a multi-period equilibrium problem with equilibrium constraints (EPEC) to investigate ESS price-makers' strategic behavior and analyze their impact on the market clearing implementation. Shahmohammadi et al. [23] proposed a novel multi-period EPEC that is linearized and solved as an MILP, examining and attempting to mitigate price-suppression effect of renewables. Two distinct EPECs were developed by Wang et al. [24], in order to model the optimal coordination of day-ahead electricity and natural gas markets. The models differ from each other on the coupling information exchanging channels and constitute a potentially useful future guide for market operators and regulators. Furthermore, Tsimopoulos and Georgiadis [25] developed a fundamental stochastic bi-level optimization approach, in the form of an EPEC model, in order to detect pure-strategy Nash equilibria in a pool-based electricity market, characterized by the assertive integration of wind power generation. The proposed methodology addressed several cases, including four different market competition conditions and network congestions.

The incertitude induced by the intermittent technologies' osmosis in power systems has reinforced the importance of ancillary services, such as reserves, as a tradable commodity, which guarantees the resilience of electricity markets. González et al. [26] reviewed the main energy and reserve market structures providing in parallel, a mathematical framework to model the majority of them. Drury et al. [27] focused on evaluating the financial gains and comparing the operational characteristics of various electricity storage technologies, such as diabatic, adiabatic and isothermal Compressed Air Energy Storage (CAES), when participating in energy and reserve markets. Reddy, Bijwe and Abhyankar [28] recommended two different energy and spinning reserve market clearing mechanisms for a wind-thermal power system under wind power generation uncertainty. Nojavan et al. [29] formulated a robust MILP to generate the optimal offering and bidding strategies for a merchant CAES in deregulated day-ahead electricity markets. Tsimopoulos and Georgiadis [30] presented a bi-level complementarity model, to determine the optimal capacity withholding strategies for a conventional-wind generation portfolio, in a jointly cleared energy and reserve market.

Several references cover applications and participation of strategic electricity storage systems in jointly cleared energy and reserve markets. More specifically, Vespermann, Delikaraoglou and Pinson [31] formulated a bi-level optimization problem to analyze the participation of a price-maker ESS operator in the day-ahead and real-time market. The procurement of upward and downward reserve capacities and their corresponding costs, over a specified planning horizon are taken into account. Nasrolahpour et al. [32] developed a bi-level complementarity model for a strategic ESS operator, participating in a sequential two-stage clearing settlement, comprised of joint day-ahead and reserve market and energy-only balancing market, under net load uncertainty. The problem is reformulated into an MILP, using appropriate Karush-Kuhn-Tucker (KKT) conditions and the binary expansion approach proposed by Ruiz, Conejo and Gabriel [33].

The bi-level optimization problem proposed by Schillemans et al. [34], also studies the participation of a price-maker ESS agent in joint energy and reserve market, though regarding a simultaneous market clearing of day-ahead and balancing stage. Khojasteh et al. [35] proposed a robust optimization approach to investigate the optimal aggregated offering strategy of the coordinated operation of wind power resources and battery energy storage systems in a joint energy and reserve market. Khaloie et al. [36] introduced a multi-objective mathematical framework to define the optimal offering strategy of a WTES (wind-thermal-storage) system in the day-ahead and spinning reserve markets. The suggested MIP problem aims at maximizing WTES's expected profit and minimizing emissions originating from thermal units, while uncertainties arose from wind power output or day-ahead and reserve market prices, are modelled through a set of scenarios.

Inspired by the market clearing optimization algorithm presented in Ref. [32], this works proposes an MPEC approach, to derive optimal offering strategies for a strategic ESS operator, participating in a sequential joint energy and reserve day-ahead and real-time market clearing settlement. While a common assumption suggests that ESS agents act as price-takers, large-scale investor-owned storage systems can offer/bid strategically in electricity markets, especially as wind power generation uncertainty increases [37]. Electricity market of California constitutes a great example of such a market environment, as California Public Utilities Commission has called the three large independently-operated utility companies to install a capacity of 1325 MW of

energy storage by 2020 [38]. Therefore, examining and analyzing the optimal strategic behavior of large-scale investor-owned energy storage systems appears to attract significant interest in recent years.

The developed bi-level mathematical framework, contrary to the formulation in Ref. [32] which regards net load uncertainty and a single-bus market scheme, introduces wind power generation uncertainty at the balancing market and incorporates network transmission constraints. This consideration introduces extra model complexity and provides strategic producer with further arbitrage opportunities, due to the transmission lines' congestions. Comparing to the proposed optimization problem at [31], this work adopts a different reserve, day-ahead and real-time pool clearing approximation and power system structure, embedding in multiple busses, incurred by network transmission constraints. In comparison to the configuration in Ref. [34] where day-ahead and balancing pools are cleared simultaneously, the proposed MPEC presents a sequential day-ahead and balancing market clearing structure, with a distinct elaboration of ESS's operational constraints, while also introducing stochastic wind power generation scenarios and examining possible line congestions, in a multiple buses settlement. Finally, the problem formulation in Ref. [36] consists of an MIP mathematical framework, which accounts for a WTES company seeking to maximize its profit and minimize expected emissions, while emission market price is treated as parameter. In this work, a non-linear bi-level model is proposed, aiming to maximizing ESS agent's profit in a multi-bus power system and clear the day-ahead and real-time market, deriving energy and reserve market prices and analyzing possible network congestions.

This work aims at providing a fundamental and systematic optimization-based framework for analyzing the strategic decisions of an ESS agent in energy and reserve markets, under wind generation uncertainty. Other types of renewable energy sources could also be considered in the mathematical framework, but this will inevitably lead to computationally intractable problems. It is important to emphasize that ESS agent is considered as profit-making agent and its detailed operational scheduling is beyond the scope of this work.

Considering the above literature review of the state-of-the-art, the main objectives and contributions of this work are as follows:

- A novel bi-level complementarity model to analyze the interrelationships between the ESS's optimal bidding and offering strategies, participating in a jointly cleared energy and reserve day-ahead pool, as well as in real-time pool, under network transmission constraints.
- Transformation of the bi-level model into an MPEC and further reduce it to the equivalent MILP model, by deploying KKT optimality conditions, strong duality theorem and a binary expansion method.
- A systematic methodology, in order to generate day-ahead, upward/downward reserve and real-time market prices, as dual variables of the corresponding energy balance constraints.
- Introduce a comprehensive market operations structure and a thorough mathematical framework for the integration of ESS in a transmission constrained multi-bus network, under wind power generation uncertainty.
- Analyze the market outcomes of transmission lines congestions and variable levels of wind power generation insertion and the measure of their influence on ESS agent's (dis)charging decision strategies.

The rest of this article is organized as follows: *Section 2* presents the bi-level mathematical formulation, as well as the adopted solution strategy. *Section 3* illustrates the applicability of the proposed framework in a 6-bus power system and analyzes the results, while summarizes computational issues. Finally, *Section 4* draws up concluding remarks and highlights potential future directions.

2. Mathematical model

The proposed bi-level programming problem, is formulated on the basis of a single leader-follower Stackelberg hypothesis' game [39] and considers the optimal offering/bidding strategies and (dis) charging decisions for a price-maker ESS operator, while competing conventional and wind generation agents, which act as price-takers in energy and reserve pools. The adopted market setup is aligned with the Pennsylvania-New Jersey-Maryland (PJM) market's structure [32,40], where wind power generation uncertainty is addressed through the real-time realization of ancillary services' operations [41].

The upper level of the proposed mathematical framework contains strategic generator's expected profit maximization problem and relies on the marginal prices, intrinsically provided by the lower-level problem's implementation. The lower-level problem on the other hand, represents the market clearing mechanism conducted by the ISO. It also determines the optimal generators' energy dispatch and reserve procurement, for the purpose of maximizing social welfare, by virtue of a two-stage stochastic program [42]. The first stage pertains to the optimization of a jointly encountered energy and upward/downward reserve day-ahead pool, deriving the optimal amounts of scheduled generation and reserve procurement, as well as their reciprocal clearing prices. At the second stage, the ISO's real-time market clearing procedure is presented, determining the deployment levels of the reserves -formerly procured in the day-ahead clearance phase- and generating the RT market prices [43]. Wind power generation uncertainty is modelled through the establishment of conceivable wind generation scenarios [44]. Considering the lower problem satisfies continuity and convexity prerequisites, the proposed bi-level problem can be reformulated to an MPEC, on the strength of the KKT optimality conditions' application. Consequently, applying Fortuny-Amat and McCarl linearization approach [45], strong duality theorem and a binary expansion method [46], the mathematical model is further reduced into an equivalent MILP, which entails reasonable computational requirements by commercial solvers such as GAMS/CPLEX [47].

2.1. Bi-level model structure

The following bi-level mathematical model is formulated to determine optimal offering and (dis)charging strategies for an ESS agent, in a joint energy and reserve market settlement, in order to counteract occasional system imbalances, imposed by wind power generation deviations and plausible network congestions (see Fig. 1).

2.1.1. Upper-level problem: ESS's expected profit maximization

The upper-level problem maximizes the expected profit for the ESS operator, stemming from its involvement in all different types of markets, and is described below by constraints (1)–(18):

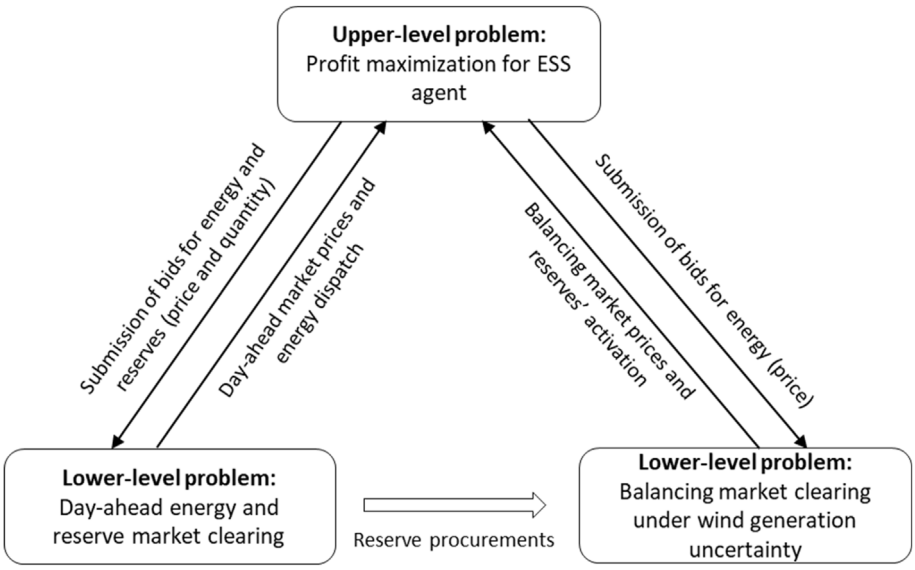


Fig. 1. Mathematical structure and function of the bi-level model.

$$\textbf{Maximize} \sum_t \sum_s \Bigg[-\left(\lambda_{n,t}^{DA} + c_s^{ch}\right)\cdot G_{s,t}^{DA,ch} + \left(\lambda_{n,t}^{DA} - c_s^{dis}\right)\cdot G_{s,t}^{DA,dis} + \left[\lambda_t^{\uparrow}\cdot\left(rp_{s,t}^{ch,\uparrow} + rp_{s,t}^{dis,\uparrow}\right) + \lambda_t^{\downarrow}\cdot\left(rp_{s,t}^{ch,\downarrow} + rp_{s,t}^{dis,\downarrow}\right)\right] + \sum_\omega \pi_\omega \cdot \Bigg[\left(\lambda_{n,t,\omega}^{RT} + c_s^{ch}\right)\cdot\left(ra_{s,t,\omega}^{ch,\uparrow} - ra_{s,t,\omega}^{ch,\downarrow}\right) + \left(\lambda_{n,t,\omega}^{RT} - c_s^{dis}\right)\cdot\left(ra_{s,t,\omega}^{dis,\uparrow} - ra_{s,t,\omega}^{dis,\downarrow}\right)\Bigg]\Bigg] \tag{1}$$

s.t.

$$x_{s,t}^{ch},\ x_{s,t}^{dis} \in \{0,1\} \quad \forall s,\ \forall t \tag{2}$$

$$x_{s,t}^{ch} + x_{s,t}^{dis} \le 1\ \forall s,\ \forall t \tag{3}$$

$$0 \le \overline{G}_{s,t}^{dis} \le x_{s,t}^{dis}\cdot G_s^{dis,\max} \quad \forall s,\ \forall t \tag{4}$$

$$0 \le \overline{rc}_{s,t}^{dis,\uparrow} \le x_{s,t}^{dis}\cdot RC_s^{dis,\uparrow,max} \quad \forall s,\ \forall t \tag{5}$$

$$0 \le \overline{rc}_{s,t}^{dis,\downarrow} \le x_{s,t}^{dis}\cdot RC_s^{dis,\downarrow,max} \quad \forall s,\ \forall t \tag{6}$$

$$\overline{G}_{s,t}^{dis} + \overline{rc}_{s,t}^{dis,\uparrow} \le x_{s,t}^{dis}\cdot G_s^{dis,max} \quad \forall s,\ \forall t \tag{7}$$

$$\overline{rc}_{s,t}^{dis,\downarrow} - \overline{G}_{s,t}^{dis} \le 0\ \forall s,\ \forall t \tag{8}$$

$$0 \le \overline{G}_{s,t}^{ch} \le x_{s,t}^{ch}\cdot G_s^{ch,max} \quad \forall s,\ \forall t \tag{9}$$

$$0 \le \overline{rc}_{s,t}^{ch,\uparrow} \le x_{s,t}^{ch}\cdot RC_s^{ch,\uparrow,max} \quad \forall s,\ \forall t \tag{10}$$

$$0 \le \overline{rc}_{s,t}^{ch,\downarrow} \le x_{s,t}^{ch}\cdot RC_s^{ch,\downarrow,max} \quad \forall s,\ \forall t \tag{11}$$

$$\overline{G}_{s,t}^{ch} + \overline{rc}_{s,t}^{ch,\downarrow} \le x_{s,t}^{ch}\cdot G_s^{ch,max} \quad \forall s,\ \forall t \tag{12}$$

$$\overline{rc}_{s,t}^{ch,\uparrow} - \overline{G}_{s,t}^{ch} \le 0\ \forall s,\ \forall t \tag{13}$$

$$O_{s,t}^{ch},\ O_{s,t}^{ch,\uparrow},\ O_{s,t}^{ch,\downarrow} \ge 0\ \forall s,\ \forall t \tag{14}$$

$$O_{s,t}^{dis},\ O_{s,t}^{dis,\uparrow},\ O_{s,t}^{dis,\downarrow} \ge 0\ \forall s,\ \forall t \tag{15}$$

$$0 \le soc \le SOC_s^{max} \quad \forall s,\ \forall t \tag{16}$$

$$soc_{s,t} = SOC_s^{ini}\ \forall s,\ t = N_t \tag{17}$$

$$soc_{s,t} = SOC_s^{ini} + \sum_k \eta^{ch}\cdot\left[G_{s,t}^{DA,ch} + \sum_\omega \pi_\omega\cdot\left(-ra_{s,t,\omega}^{ch,\uparrow} + ra_{s,t,\omega}^{ch,\downarrow}\right)\right] - \sum_k \frac{1}{\eta^{dis}}\cdot\left[G_{s,t}^{DA,dis} + \sum_\omega \pi_\omega\cdot\left(ra_{s,t,\omega}^{dis,\uparrow} - ra_{s,t,\omega}^{ch,\downarrow}\right)\right] \forall s, \forall t \tag{18}$$

Objective function (1) includes all ESS operator's revenue and cost terms, generated from its participation in the day-ahead, reserve and balancing pool. More specifically, the first line

corresponds to the day-ahead market profit, derived from the energy discharged income, $\lambda^{DA}_{n,t} \cdot G^{DA,dis}_{s,t}$, operational charging/discharging costs, $c^{ch}_s \cdot G^{DA,ch}_{s,t} / c^{dis}_s \cdot G^{DA,dis}_{s,t}$, as well as the cost for purchasing energy during charging periods, $\lambda^{DA}_{n,t} \cdot G^{DA,ch}_{s,t}$. The second line, refers to the ESS's income, originating from the upward and downward reserve procurement, in the course of day-ahead market clearing. The remaining lines, correlate with the ESS's profit in the balancing market. The upward charging reserve activation, $ra^{ch,\uparrow}_{s,t,\omega}$ constitutes an income factor for the storage system agent, as it is compensated from the market at real-time price, $\lambda^{RT}_{n,t,\omega}$, for its charging power curtailment and is exempt from the corresponding operational charging cost, c^{ch}_s. Regarding downward charging reserve activation, the storage system disburses funds to purchase the additional charging power, $\lambda^{RT}_{n,t,\omega} \cdot ra^{ch,\downarrow}_{s,t,\omega}$, and to cover the necessary operational costs, $c^{ch}_s \cdot ra^{ch,\downarrow}_{s,t,\omega}$. In the upward discharging reserve activation mode, the energy storage agent provides power to the market, acquiring earnings based on the real-time price, $\lambda^{RT}_{n,t,\omega} \cdot ra^{dis,\uparrow}_{s,t,\omega}$ and bearing the cost of discharging function, $c^{dis}_s \cdot ra^{dis,\uparrow}_{s,t,\omega}$. Conclusively, in the downward discharging reserve activation mode, ESS agent compensates market for the energy not dispatched $\lambda^{RT}_{n,t,\omega} \cdot ra^{dis,\downarrow}_{s,t,\omega}$, while freed from discharging operational expenses, $c^{dis}_s \cdot ra^{dis,\downarrow}_{s,t,\omega}$.

Constraints (2) and (3) declare the binary nature and ensure the mutually exclusive function of the charging/discharging decision variables. Constraints (4)–(6) and (9)–(11) limit the energy and reserve capacity offers/bids of the ESS, in discharging and charging mode correspondingly. Constraints (7), (8) and (12), (13) ensure that the summation of energy and reserve quantity offers/bids, cannot exceed the discharging/charging maximum capacity of the storage system. Furthermore, constraints (14), (15) guarantee the non-negative interpretation of energy and reserve bidding/offering prices, whereas constraints (16) and (17) enforce that the state of charge at the end of the planning horizon harmonizes with the initial one. Constraints (18) represent the ESS's hourly state of charge fluctuations, depending on the ESS's optimal operational decisions, with η^{ch}, η^{dis} denoting charging and discharging efficiencies, respectively. The analyzed upper-level problem is constrained by the two-stage lower-level problem (19)-(44) and (45)-(59), described in the following sections.

2.1.2. Lower-level problem: joint energy and reserve day-ahead market clearing

The lower-level problem thoroughly investigates a two-stage sequentially cleared day-ahead and real-time market structure, providing the optimal energy dispatch and reserve procurement, as well as deriving marginal prices, as dual variables of the energy and reserve equilibriums analogously. It is critical to mention that ESS agent's offering/bidding decisions $O^{ch}_{s,t}$, $O^{ch,\uparrow}_{s,t}$, $O^{ch,\downarrow}_{s,t}$, $O^{dis}_{s,t}$, $O^{dis,\uparrow}_{s,t}$, $O^{dis,\downarrow}_{s,t}$ are treated as variables in the profit maximization upper-level problem, while in the lower-level problem are encountered by ISO as parameters. Hence, the mathematical framework presented in *Section 2.2.2* and *2.2.3* is classified as linear and convex [9].

$$\begin{aligned}\textbf{Maximize}\ & \sum_s \left(O^{ch}_{s,t} \cdot G^{DA,ch}_{s,t} - O^{dis}_{s,t} \cdot G^{DA,dis}_{s,t}\right) + \sum_d u_{d,t} \cdot L^{DA}_{d,t} \\ & - \sum_i c_{i,t} \cdot P^{DA}_{i,t} - \sum_j c^{w}_{j,t} \cdot W^{DA}_{j,t} \\ & - \sum_s \left(O^{ch,\uparrow}_{s,t} \cdot rp^{ch,\uparrow}_{s,t} + O^{ch,\downarrow}_{s,t} \cdot rp^{ch,\downarrow}_{s,t} + O^{dis,\uparrow}_{s,t} \cdot rp^{dis,\uparrow}_{s,t} + O^{dis,\downarrow}_{s,t} \cdot rp^{dis,\downarrow}_{s,t}\right) \\ & - \sum_i c^{res}_{i,t} \cdot \left(rpc^{\uparrow}_{i,t} + rpc^{\downarrow}_{i,t}\right) \\ & - \sum_d u^{res}_{d,t} \cdot \left(rpd^{\uparrow}_{d,t} + rpd^{\downarrow}_{d,t}\right) \quad \forall t \end{aligned} \tag{19}$$

s.t.

$$\sum_{d \in DaN} L^{DA}_{d,t} - \sum_{s \in SaN} \left(G^{DA,dis}_{s,t} - G^{DA,ch}_{s,t}\right) - \sum_{i \in IaN} P^{DA}_{i,t} - \sum_{j \in JaN} W^{DA}_{j,t} + \sum_{m \in NaM} B_{n,m} \cdot \left(\delta^{\circ}_{n,t} - \delta^{\circ}_{m,t}\right) = 0 \quad : \left[\lambda^{DA}_{n,t}\right] \quad \forall n, \forall t \tag{20}$$

$$\sum_s \left(rp^{ch,\uparrow}_{s,t} + rp^{dis,\uparrow}_{s,t}\right) + \sum_i rpc^{\uparrow}_{i,t} + \sum_d rpd^{\uparrow}_{d,t} = R^{\uparrow}_t \quad : \left[\lambda^{\uparrow}_t\right] \quad \forall t \tag{21}$$

$$\sum_s \left(rp^{ch,\downarrow}_{s,t} + rp^{dis,\downarrow}_{s,t}\right) + \sum_i rpc^{\downarrow}_{i,t} + \sum_d rpd^{\downarrow}_{d,t} = R^{\downarrow}_t \quad : \left[\lambda^{\downarrow}_t\right] \quad \forall t \tag{22}$$

$$0 \le P^{DA}_{i,t} \le P^{max}_i \quad : \left[a^{min}_{i,t}, a^{max}_{i,t}\right] \quad \forall i,\ \forall t \tag{23}$$

$$0 \le L^{DA}_{d,t} \le L^{max}_{d,t} \quad : \left[\beta^{min}_{d,t}, \beta^{max}_{d,t}\right] \quad \forall d,\ \forall t \tag{24}$$

$$0 \le G^{DA,ch}_{s,t} \le \overline{G}^{ch}_{s,t} \quad : \left[\gamma^{ch,min}_{s,t}, \gamma^{ch,max}_{s,t}\right] \quad \forall s,\ \forall t \tag{25}$$

$$0 \le G^{DA,dis}_{s,t} \le \overline{G}^{dis}_{s,t} \quad : \left[\gamma^{dis,min}_{s,t}, \gamma^{dis,max}_{s,t}\right] \quad \forall s,\ \forall t \tag{26}$$

$$0 \le rp^{ch,\uparrow}_{s,t} \le \overline{rc}^{ch,\uparrow}_{s,t} \quad : \left[\gamma^{ch,\uparrow,min}_{s,t}, \gamma^{ch,\uparrow,max}_{s,t}\right] \quad \forall s,\ \forall t \tag{27}$$

$$0 \le rp_{s,t}^{ch,\downarrow} \le \overline{rc}_{s,t}^{ch,\downarrow} \quad : \left[\gamma_{s,t}^{ch,\downarrow,min}, \gamma_{s,t}^{ch,\downarrow,max}\right] \quad \forall s,\ \forall t \tag{28}$$

$$0 \le rp_{s,t}^{dis,\uparrow} \le \overline{rc}_{s,t}^{dis,\uparrow} \quad : \left[\gamma_{s,t}^{dis,\uparrow,min}, \gamma_{s,t}^{dis,\uparrow,max}\right] \quad \forall s,\ \forall t \tag{29}$$

$$0 \le rp_{s,t}^{dis,\downarrow} \le \overline{rc}_{s,t}^{dis,\downarrow} \quad : \left[\gamma_{s,t}^{dis,\downarrow,min}, \gamma_{s,t}^{dis,\downarrow,max}\right] \quad \forall s,\ \forall t \tag{30}$$

$$rp_{s,t}^{ch,\uparrow} \le G_{s,t}^{DA,ch} \quad : \left[\gamma_{s,t}^{ch,\uparrow}\right] \quad \forall s,\ \forall t \tag{31}$$

$$rp_{s,t}^{dis,\downarrow} \le G_{s,t}^{DA,dis} \quad : \left[\gamma_{s,t}^{dis,\downarrow}\right] \quad \forall s,\ \forall t \tag{32}$$

$$0 \le rpc_{i,t}^{\uparrow} \le RCc_i^{\uparrow,max} \quad : \left[a_{i,t}^{\uparrow,min}, a_{i,t}^{\uparrow,max}\right] \quad \forall i,\ \forall t \tag{33}$$

$$0 \le rpc_{i,t}^{\downarrow} \le RCc_i^{\downarrow,max} \quad : \left[a_{i,t}^{\downarrow,min}, a_{i,t}^{\downarrow,max}\right] \quad \forall i,\ \forall t \tag{34}$$

$$P_{i,t}^{DA} + rpc_{i,t}^{\uparrow} \le P_i^{max} \quad : \left[a_{i,t}^{\uparrow}\right] \quad \forall i,\ \forall t \tag{35}$$

$$rpc_{i,t}^{\downarrow} - P_{i,t}^{DA} \le 0 \quad : \left[a_{i,t}^{\downarrow}\right] \quad \forall i,\ \forall t \tag{36}$$

$$0 \le rpd_{d,t}^{\uparrow} \le RCd_{d,t}^{\uparrow,max} \quad : \left[\beta_{d,t}^{\uparrow,min}, \beta_{d,t}^{\uparrow,max}\right] \quad \forall d,\ \forall t \tag{37}$$

$$0 \le rpd_{d,t}^{\downarrow} \le RCd_{d,t}^{\downarrow,max} \quad : \left[\beta_{d,t}^{\downarrow,min}, \beta_{d,t}^{\downarrow,max}\right] \quad \forall d,\ \forall t \tag{38}$$

$$L_{d,t}^{DA} + rpd_{d,t}^{\downarrow} \le L_{d,t}^{max} \quad : \left[\beta_{i,t}^{\downarrow}\right] \quad \forall d,\ \forall t \tag{39}$$

$$rpd_{d,t}^{\uparrow} - L_{d,t}^{DA} \le 0 \quad : \left[\beta_{i,t}^{\uparrow}\right] \quad \forall d,\ \forall t \tag{40}$$

$$0 \le W_{j,t}^{DA} \le W_{j,t}^{max} \quad : \left[\varepsilon_{j,t}^{min}, \varepsilon_{j,t}^{max}\right] \quad \forall j,\ \forall t \tag{41}$$

$$-T_{n,m}^{max} \le B_{n,m} \cdot \left(\delta_{n,t}^{\circ} - \delta_{m,t}^{\circ}\right) \le T_{n,m}^{max} \quad : \left[\zeta_{n,m,t}^{min}, \zeta_{n,m,t}^{max}\right] \ \forall n, \forall m \in NaM, \forall t \tag{42}$$

$$-\pi \le \delta_{n,t}^{\circ} \le \pi \quad : \left[\tilde{\zeta}_{n,t}^{min}, \tilde{\zeta}_{n,t}^{max}\right] \quad \forall n, \forall t \tag{43}$$

$$\delta_{n_1,t}^{\circ} = 0 \quad : \left[\zeta_{n,t}^{\circ}\right] \ \forall n = n_1, \forall t \tag{44}$$

Objective function (19) constitutes the basis for the first stage optimization and represents the joint day-ahead and reserve market clearing, maximizing total social welfare. Alternative definitions for the objective function can be considered the economic dispatch or total social cost minimization, given the inelasticity of electricity demand. It is worth mentioning that conventional generators, energy storage systems and demands are capable of procuring both upward and downward reserves, in order to contribute to market's function stability, in contrast to wind power generation systems, whose erratic nature excludes equivalent potential. Constraints (20), (21) and (22) ensure energy and upward/downward reserve procurement equilibriums, deriving day-ahead market's energy and reserve prices, as their corresponding dual variables. Constraints (23) and (24) limit conventional generators' production and demands' consumption level, respectively. Constraints (25) and (26) restrict ESS's charging/discharging energy levels, while (27)–(32) restrain upward/downward (dis)charging reserve provision. Equally, constraints (33)–(36) and (37)–(40) limit conventional generators' and demand's reserve procurements, respectively. Constraints (41) on the other hand, circumscribe the wind power dispatch levels in the day-ahead market. Finally, constraints (42) enforce capacity limits for network transmission lines, while constraints (43), (44) limit each bus's voltage angle range and define $n1$ as power grid's slack bus, at the day-ahead market.

2.1.3. Lower-level problem: real-time market clearing under wind power generation uncertainty

The second stage of the lower-level problem represents real-time market clearing under stochastic wind power generation. The participation of non-dispatchable generators, such as wind, into the power grid can evoke imbalances into the final supply-demand equilibrium, due to their volatile electricity provision. Thus, the existence of the balancing market accounting for these supply deviations, can contribute establishing an auxiliary stability tier.

$$\begin{aligned} \textbf{Minimize} \quad & \sum_d VOLL_{d,t} \cdot L_{d,t,\omega}^{sh} + \sum_j c_{j,t}^{w} \cdot \left(W_{j,t,\omega}^{RT} - W_{j,t}^{DA} - W_{j,t,\omega}^{SP}\right) \\ & + \sum_i c_{i,t} \cdot \left(rac_{i,t,\omega}^{\uparrow} - rac_{i,t,\omega}^{\downarrow}\right) \\ & + \sum_s O_{s,t}^{dis} \cdot \left(ra_{s,t,\omega}^{dis,\uparrow} - ra_{s,t,\omega}^{dis,\downarrow}\right) \\ & + \sum_s O_{s,t}^{ch} \cdot \left(ra_{s,t,\omega}^{ch,\uparrow} - ra_{s,t,\omega}^{ch,\downarrow}\right) \\ & + \sum_d u_{d,t} \cdot \left(rad_{d,t,\omega}^{\uparrow} - rad_{d,t,\omega}^{\downarrow}\right) \quad \forall t, \omega \end{aligned} \tag{45}$$

s.t.

$$\begin{aligned} & \sum_{d \in DaN} L_{d,t,\omega}^{sh} + \sum_{s \in SaN} \left(ra_{s,t,\omega}^{ch,\uparrow} - ra_{s,t,\omega}^{ch,\downarrow}\right) + \sum_{s \in SaN} \left(ra_{s,t,\omega}^{dis,\uparrow} - ra_{s,t,\omega}^{dis,\downarrow}\right) \\ & \quad + \sum_{i \in IaN} \left(rac_{i,t,\omega}^{\uparrow} - rac_{i,t,\omega}^{\downarrow}\right) + \sum_{d \in DaN} \left(rad_{d,t,\omega}^{\uparrow} - rad_{d,t,\omega}^{\downarrow}\right) \\ & \quad + \sum_{j \in JaN} \left(W_{j,t,\omega}^{RT} - W_{j,t}^{DA} - W_{j,t,\omega}^{SP}\right) \\ & \quad - \sum_{m \in NaM} B_{n,m} \cdot \left(\delta_{n,t,\omega} - \delta_{n,t}^{\circ} + \delta_{m,t}^{\circ} - \delta_{m,t,\omega}\right) = 0 \\ & \quad : \left[\lambda_{n,t,\omega}^{RT}\right] \quad \forall n, \forall t, \forall \omega \end{aligned} \tag{46}$$

$$0 \le rac_{i,t,\omega}^{\uparrow} \le rpc_{i,t}^{\uparrow} \quad : \left[\theta_{i,t,\omega}^{\uparrow,min}, \theta_{i,t,\omega}^{\uparrow,max}\right] \quad \forall i, \forall t, \forall \omega \tag{47}$$

$$0 \le rac_{i,t,\omega}^{\downarrow} \le rpc_{i,t}^{\downarrow} \quad : \left[\theta_{i,t,\omega}^{\downarrow,min}, \theta_{i,t,\omega}^{\downarrow,max}\right] \quad \forall i, \forall t, \forall \omega \tag{48}$$

$$0 \le rad_{d,t,\omega}^{\uparrow} \le rpd_{d,t}^{\uparrow} \quad : \left[\mu_{d,t,\omega}^{\uparrow,min}, \mu_{d,t,\omega}^{\uparrow,max}\right] \quad \forall d, \forall t, \forall \omega \tag{49}$$

$$0 \le rad_{d,t,\omega}^{\downarrow} \le rpd_{d,t}^{\downarrow} \quad : \left[\mu_{d,t,\omega}^{\downarrow,min}, \mu_{d,t,\omega}^{\downarrow,max}\right] \quad \forall d, \forall t, \forall \omega \tag{50}$$

$$0 \le ra_{s,t,\omega}^{dis,\uparrow} \le rp_{s,t}^{dis,\uparrow} \quad : \left[\nu_{s,t,\omega}^{dis,\uparrow,min}, \nu_{s,t,\omega}^{dis,\uparrow,max}\right] \quad \forall s, \forall t, \forall \omega \tag{51}$$

$$0 \le ra_{s,t,\omega}^{dis,\downarrow} \le rp_{s,t}^{dis,\downarrow} \quad : \left[\nu_{s,t,\omega}^{dis,\downarrow,min}, \nu_{s,t,\omega}^{dis,\downarrow,max}\right] \quad \forall s, \forall t, \forall \omega \tag{52}$$

$$0 \le ra_{s,t,\omega}^{ch,\uparrow} \le rp_{s,t}^{ch,\uparrow} \quad : \left[\nu_{s,t,\omega}^{ch,\uparrow,min}, \nu_{s,t,\omega}^{ch,\uparrow,max}\right] \quad \forall s, \forall t, \forall \omega \tag{53}$$

$$0 \le ra_{s,t,\omega}^{ch,\downarrow} \le rp_{s,t}^{ch,\downarrow} \quad : \left[\nu_{s,t,\omega}^{ch,\downarrow,min}, \nu_{s,t,\omega}^{ch,\downarrow,max}\right] \quad \forall s, \forall t, \forall \omega \tag{54}$$

$$0 \le L_{d,t,\omega}^{sh} \le L_{d,t}^{DA} \quad : \left[\mu_{d,t,\omega}^{min}, \mu_{d,t,\omega}^{max}\right] \quad \forall d, \forall t, \forall \omega \tag{55}$$

$$0 \le W_{j,t,\omega}^{sp} \le W_{j,t,\omega}^{RT} \quad : \left[\xi_{j,t,\omega}^{min}, \xi_{j,t,\omega}^{max}\right] \quad \forall j, \forall t, \forall \omega \tag{56}$$

$$-T_{n,m}^{max} \le B_{n,m} \cdot \left(\delta_{n,t,\omega} - \delta_{m,t,\omega}\right) \le T_{n,m}^{max} \quad : \left[\varphi_{n,m,t,\omega}^{min}, \varphi_{n,m,t,\omega}^{max}\right] \quad \forall n, \forall m \in NaM, \forall t, \forall \omega \tag{57}$$

$$-\pi \le \delta_{n,t,\omega} \le \pi \quad : \left[\tilde{\varphi}_{n,t,\omega}^{min}, \tilde{\varphi}_{n,t,\omega}^{max}\right] \quad \forall n, \forall t, \forall \omega \tag{58}$$

$$\delta_{n_1,t,\omega} = 0 \quad : \left[\varphi_{n,t,\omega}^{\circ}\right] \quad \forall n = n_1, \forall t, \forall \omega \tag{59}$$

Objective function (45) portrays the minimization of the expected social imbalance cost of the system, conducted by ISO and contains the following terms: i) load shedding cost, ii) wind power spillage cost iii) conventional generators' upward/downward reserve activation savings/cost iv) ESS's upward/downward (dis) charging reserve activation savings/cost, v) demands' upward/downward reserve activation savings/cost. Constraints (46) on the other hand, constitute the real-time energy balances and counteract the supply-demand equilibrium's asymmetries, posed by wind production discontinuities in the multi-bus network. Constraints (47)–(54) regulate activated energy's levels, with respect to the equivalent amounts, already procured in the day-ahead market. Concerning constraints (55) and (56), they ensure that both curtailed load and wind spillage cannot exceed day-ahead energy consumption and real-time wind production correspondingly. Conclusively, constraints (57)–(59) apply limits to the network's transmission lines capacity and buses' voltage angle, while establishing $n1$ as the real-time market slack bus.

2.2. Solution approach

The solution approach aims at reformulating the above analyzed bi-level model, into a computationally solvable single-level model, while simultaneously eradicating any inherent non-linearities. Given the linear and continuous nature of lower-level problems *2.2.2* and *2.2.3*, KKT first order optimality conditions [9] can be applied, thus recasting the bi-level optimization problem, into a single level MPEC. Due to manuscript space limitations, the KKT optimality conditions of the lower-level problems and all the necessary mathematical reformulations, are detailed in the Supplementary material.

The resulted non-linear complementarity conditions (S.15)-(S.50) and (S.62)-(S.85) of the form $0 \le g(x) \perp \mu \ge 0$, are replaced by Eqs. (S.86)-(S.158) and (S.159)-(S.206) respectively, using the Fortuny-Amat and McCarl big-M linearization technique [45].

The determination of big-Ms comprises a significant challenge, as both the over- and underestimation of their value can lead to an unrestricted or computationally and numerically inadequate problem, respectively. While many algorithmic approaches have been developed for this determination, in this work the value of Ms was established at 10^4, after implementing a series of trials, aiming at achieving an optimal solution time and results' robustness equilibrium. Strong duality theorem is applied to the day-ahead and real-time ISO's optimization problems, in order to incorporate dual variables and generate a linear equivalent for the objective functions (19) and (45) respectively. Combining these formulations, objective function (1) can be recast into the following expression:

$$\begin{aligned}
\textbf{Minimize} \quad & \sum_t \sum_s \left[\left(c_s^{ch} \cdot G_{s,t}^{DA,ch} + c_s^{dis} \cdot {}^{DA,dis}_{s,t} \right) - \sum_d u_{d,t} \cdot L_{d,t}^{DA} \right. \\
& + \sum_i c_{i,t} \cdot P_{i,t}^{DA} + \sum_i c_{i,t}^{res} \cdot \left(rpc_{i,t}^{\uparrow} + rpc_{i,t}^{\downarrow} \right) \\
& + \sum_d u_{d,t}^{res} \cdot \left(rpd_{d,t}^{\uparrow} + rpd_{d,t}^{\downarrow} \right) - \Omega_t^{DA} \\
& + \sum_\omega \pi_\omega \\
& \cdot \left[-c_s^{ch} \cdot \left(ra_{s,t,\omega}^{ch,\uparrow} - ra_{s,t,\omega}^{ch,\downarrow} \right) + c_s^{dis} \cdot \left(ra_{s,t,\omega}^{dis,\uparrow} - ra_{s,t,\omega}^{dis,\downarrow} \right) \right. \\
& + \sum_d VOLL_{d,t} \cdot L_{d,t,\omega}^{sh} + \sum_i c_{i,t} \cdot \left(rac_{i,t,\omega}^{\uparrow} - rac_{i,t,\omega}^{\downarrow} \right) \\
& \left. \left. + \sum_d u_{d,t} \cdot \left(rad_{d,t,\omega}^{\uparrow} - rad_{d,t,\omega}^{\downarrow} \right) - \Omega_{t,\omega}^{RT} \right] \right]
\end{aligned} \tag{S.247}$$

The remaining non-linearities in the objective function (S.247), are due to the term $\Omega_{t,\omega}^{RT}$ (S.232), which consists of bilinear terms i.e., the product of a primal day-ahead and a dual real-time variable $\{\{\theta_{i,t,\omega}^{\uparrow,max} \cdot rpc_{i,t}^{\uparrow}, \theta_{i,t,\omega}^{\downarrow,max} \cdot rpc_{i,t}^{\downarrow}, \mu_{d,t,\omega}^{\uparrow,max} \cdot rpd_{d,t}^{\uparrow}, \mu_{d,t,\omega}^{\downarrow,max} \cdot rpd_{d,t}^{\downarrow}, \mu_{d,t,\omega}^{max} \cdot L_{d,t}^{DA}, \lambda_{n,t,\omega}^{RT} \cdot W_{j,t}^{DA}, \lambda_{n,t,\omega}^{RT} \cdot \delta_{n,t}^{\circ}\}$. The linearization process of these terms originates from a binary expansion method [32,46], as illustrated in the Supplementary material.

3. Application study

The proposed bi-level model is implemented into a 6-bus transmission constrained network and consists of two main areas, interrelated through the transmission lines that connect buses n2-n4 and n3-n6, as illustrated by Fig. 2. The left area, which consists of buses n1, n2 and n3 is characterized by augmented generation and limited consumption levels, while in the right area (buses n4, n5, n6) a diametrically opposed scheme is adopted. The reason for this network separation lies in investigating the behavior of the ESS agent and the rest of the market participants in case of network congestion. Strategic ESS agent owns storage units $s1$, $s2$, each one established at different area and incurred by unique (dis)

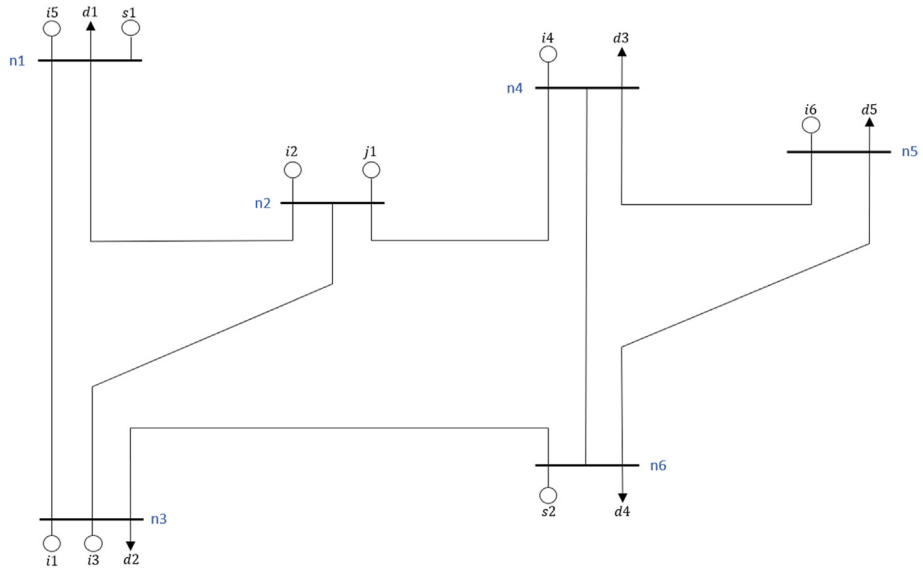


Fig. 2. 6-bus network.

Table 1
Data for ESS's units.

Units	$G_s^{ch,max}$ (MW)	c_s^{ch} $(€/MWh)$	$G_s^{dis,max}$ (MW)	c_s^{dis} $(€/MWh)$	$\frac{RC_s^{ch,\uparrow,max}}{RC_s^{ch,\downarrow,max}}$ (MW)	$\frac{RC_s^{dis,\uparrow,max}}{RC_s^{dis,\downarrow,max}}$ (MW)	SOC_s^{max} (MW)	SOC_s^{ini} (MW)
$s1$	50	4	50	24	12	12	120	0
$s2$	40	5	40	25	10	10	100	0

Table 2
Data for conventional and wind generators.

Units	P_i^{max} (MW)	$W_{j,t}^{max}$ (MW)	$RCc_i^{\uparrow,max}, RCc_i^{\downarrow,max}$ (MW)	$c_{i,t}^{res}$ $(€/MWh)$	$W_{j,t,\omega1}^{RT}, W_{j,t,\omega2}^{RT}$ (MWh)	$c_{j,t}^{w}$ $(€/MWh)$
$i1$	80	–	8,8	7	–	–
$i2$	80	–	7,7	5.5	–	–
$i3$	100	–	6,6	7	–	–
$i4$	75	–	10,10	6	–	–
$i5$	70	–	9,9	5	–	–
$i6$	100	–	9,9	7	–	–
$j1$	–	70	–	–	70,12	0

charging costs and installed energy and reserve capacity tiers, as shown in Table 1. Unit $s1$ is characterized by greater energy-reserve capacities and reduced marginal costs comparing to $s2$, therefore being more flexible and inexpensive for the ESS operator. The capacity and cost offer of each storage unit is chosen to be slightly different, considering that their installation on different buses can affect these parameters, since each bus is governed by different conditions. Conventional units $i1 - i6$ and wind power generator $j1$, represent a non-strategic producer apiece and their technical data are given in Table 2. Conventional generators $i4 - i6$ are associated with greater flexibility, as they can provide higher reserve provisions, while generators $i2, i6$ claim extended upward and downward reserve procurements, due to their inexpensive reserve offers. Wind power generation uncertainty is captured through a dual-scenario approximation i.e., a high wind generation scenario $\omega1$ and a low wind generation scenario $\omega2$, with occurrence probability 0.7 and 0.3 respectively. Conventional generators' cost offers for dispatching energy are summarized in Table S1 in the Supplementary material, in the interest of presentation purposes. It is necessary to mention that all Table numbers starting with an S are available in the Supplementary material. From this table, it can be seen that conventional producers $i1, i4, i6$ offer their energy provision at lower prices than the rest of the producers, increasing their possibility of being chosen by ISO, for dispatching energy in the day-ahead and balancing pool. The value of lost load $VOLL_{d,t}$ equals to 200 €/MWh for all the demand loads.

Furthermore, the (dis)charging efficiencies η^{ch}, η^{dis} equal to 1 for both storage units $s1$, $s2$, while maximum capacity and susceptance for each grid's transmission line $T_{n,m}^{max}$, $B_{n,m}$, equal to 200 MW and 12.412, respectively.

3.1. Uncongested network

The proposed optimization framework is solved for this case, assuming an uncongested network, using the GAMS/CPLEX solver [47]. The strategic ESS operator participates in the day-ahead market -designated as a joint energy and reserve establishment- and real-time market, through two individually operating electricity storage units $s1$, $s2$, on a 24-h horizon basis. Storage systems behave strategically by charging, when the day-ahead prices range

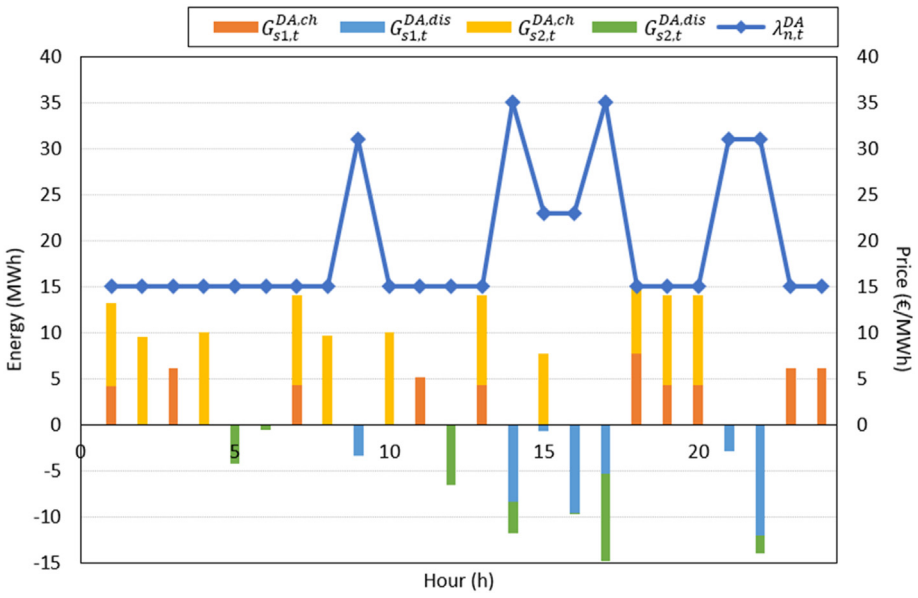


Fig. 3. Day-ahead prices and energy (dis)charging quantities for storage systems $s1, s2$.

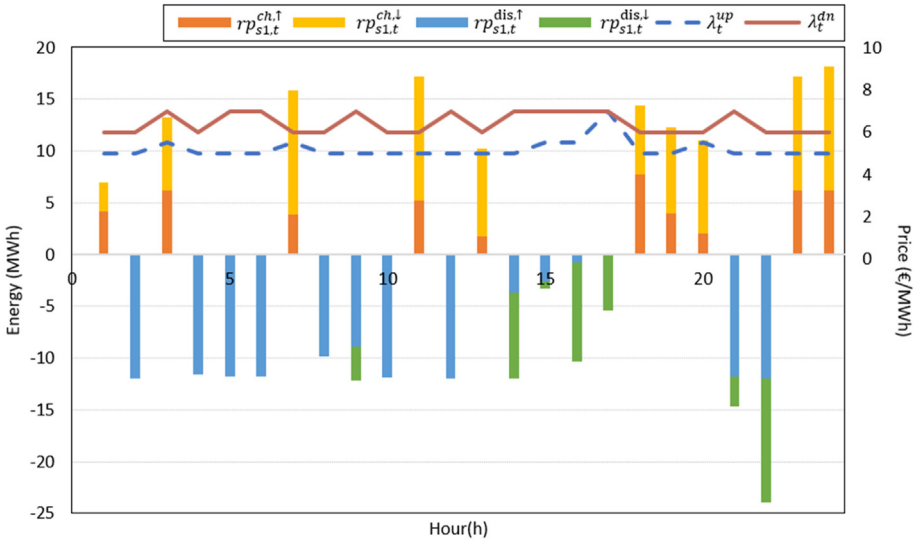


Fig. 4. Upward and downward day-ahead reserve market prices and procurements for storage system $s1$.

at low levels and discharging at time periods, identified by high prices. More specifically, as shown in Table S4, unit $s1$ strategically bids at 200 €/MWh during charging time periods (i.e., 1,3,7,11, 13, 18−20,23-24) to ensure it is being supplied, as the day-ahead prices are low. On the other hand, the same unit discharges at time periods 9, 14−17, 21−22, exploiting the elevated energy prices, and maximizing its profit share. An equivalent bidding/offering arbitrage approach is adopted by storage unit $s2$. The results are presented in Table S6 in the Supplementary material.

It is important to emphasize that, ESS's strategic decisions dictate low price offers, to ensure that the storage systems are being selected from ISO to dispatch energy in the day-ahead market, displacing the non-strategic producers from the merit-order. Concerning the reserve market realization, storage systems offer their upward and downward reserve provisions at a low price, in order to assure their robust presence in this type of market and capitalize this advantageous position in the real-time market, where the clearing prices can approximate 200 €/MWh. To ratify the above policy, Table S5 and Table S7 reveal that the storage systems $s1$ and $s2$, are actively involved in the balancing market and increase their revenues mostly by deploying upward charging and discharging reserves in the low wind generation scenario $\omega 2$.

Intending to thoroughly investigate the storage system's $s1$. Operation, its behavior is analyzed in time period 14. At this time interval, it can be noticed from Fig. 3 that the day-ahead price increases up to 35 €/MWh (due to the cost offer of marginal producer $i3$) and therefore the storage system $s1$ decides to discharge 8.3 MWh of energy. In parallel, system $s1$ strategically procures 8.3 MW of downward and 3.7 MW of upward reserve in discharging mode. As illustrated by Fig. 4, the upward and downward reserve prices are 5 €/MWh and 7 €/MWh respectively and are defined by the participation of marginal conventional generators, $i5$, $i6$, in the reserve market. Conclusively, as demonstrated in Table S5, unit $s1$ activates the entire amount of the already engaged upward and

downward reserves, depending on the real-time wind generation scenario $\omega 1$, $\omega 2$, and is compensated in the corresponding premium balancing price i.e., 41 €/MWh and 200 €/MWh.

Another, significant observation derived from Table S5 is related to the amount of activated upward reserves in discharging mode, for the real-time scenario $\omega 2$. Due to the curtailed wind power generation (i.e., 12 MWh) comparing to scenario $\omega 1$ (70 MWh), the storage system $s1$ is deployed by the Independent System Operator, for the purpose of providing additional energy and balancing the supply-demand equilibrium. Indisputably, this contribution of the storage system $s1$ in the balancing market is associated with significant income sources, as formerly mentioned. Table S7 demonstrates an homologous demeanor pursued by the energy storage system $s2$.

Apropos of the price-taker agents' involvement in the market, wind generator $j1$ is nominated by ISO to dispatch the entire amount of its electricity production in the day-ahead pool (70 MWh), for each time period, as a result of its zero cost offers' contribution to the market prices suppression. Analogous results are realized in the real-time market framework, with wind generator $j1$ dispatching his whole electricity production. As for the balancing market prices, as shown in Table S5, low wind power generation in real-time scenario $\omega 2$ (i.e., 12 MWh) stimulates explicit escalation of the market price up to 200 €/MWh, comparing to the equivalent market price generated from the high wind generation scenario $\omega 1$, which ranges between 16 €/MWh and 41€/MWh. On the other hand, conventional generators' day-ahead dispatch and reserve procurement, are determined in respect of their equivalent energy and upward/downward reserve cost offers. Specifically, as depicted in Table S8, producers $i1, i3, i4, i6$ are characterized by relatively low generation marginal costs, therefore generating energy for the whole 24-h planning horizon. Furthermore, unit $i5$ occasionally dispatch energy to the day-ahead pool, while unit $i2$ is constantly eliminated by the ISO clearing, due to its higher-priced offers. On the contrary, units $i1, i2, i5$ dominate the upward and downward reserve provision settlement -and consequently the balancing market-overthrowing the remaining conventional generators, since their correlated offering is summarized in 5.5 €/MWh, 6 €/MWh and 5 €/MWh respectively, as presented in Table 2 and Table S8. It is also worth mentioning that, equally to the ESSs' operation, dispatchable generators preeminently deploy their procured reserves in the low wind generation balancing scenario $\omega 2$, to offset possible system supply imbalances, as Table S9 renders.

Fig. 5 illustrates the storage systems' $s1$, $s2$ state of charge for an 24-h horizon. More specifically, unit's $s2$ state of charge varies at higher levels than the equivalent of the unit $s1$ and this policy can be justified by their different marginal costs in discharging mode, 24 €/MWh and 25 €/MWh, respectively. Due to this operating cost distinction, storage unit $s1$ is favored by the ESS agent, to discharge higher amounts of energy in the day-ahead market (42.19 MWh), comparing to the corresponding from the unit $s2$ (26.25 MWh), thus leading to higher energy accumulation for storage unit $s2$.

3.2. Congested network

In this case study, power systems congestions are considered for the 6-bus network, as a result of a capacity decrement for the transmission line 3–6. More precisely, while the maximum power flow between buses $n3 - n6$ for the uncongested network reaches up to 30.33 MW, the system's response during the same 24-h horizon when line capacity is reduced to 15 MW is studied. A substantial differentiation occurred, pertains to the existence of different local marginal prices (LMPs) in the day-ahead market, due to the impeded energy transfer among buses, as shown in Fig. 6.

Despite these design modifications, the main ESS agent's offering/bidding and (dis)charging principles comply with the overall strategic behavior, adopted at the uncongested network case. As Fig. 7 indicates, storage system $s1$ discharges at time periods when the day-ahead price for the bus $n1$ fluctuates at high levels, such as 33.7 €/MWh and 37 €/MWh for time periods 9,21,22 and 14,17, respectively. Equivalently, unit $s1$ charges when the price levels are reduced i.e., time periods 1–4,6-8,10–13,15-16,18, 20, 23. Fig. 7, also illustrates the upward and downward reserve procurements in the dis (charging) mode, which constitute a precursor of storage system's $s1$ involvement in the real-time market.

As already mentioned in *Section 3.1*, the network is discretized into two subareas i.e., the left one where the installed capacity outblanaces demand loads and the right one, where the potential power generation yields at the demand's higher quantity levels.

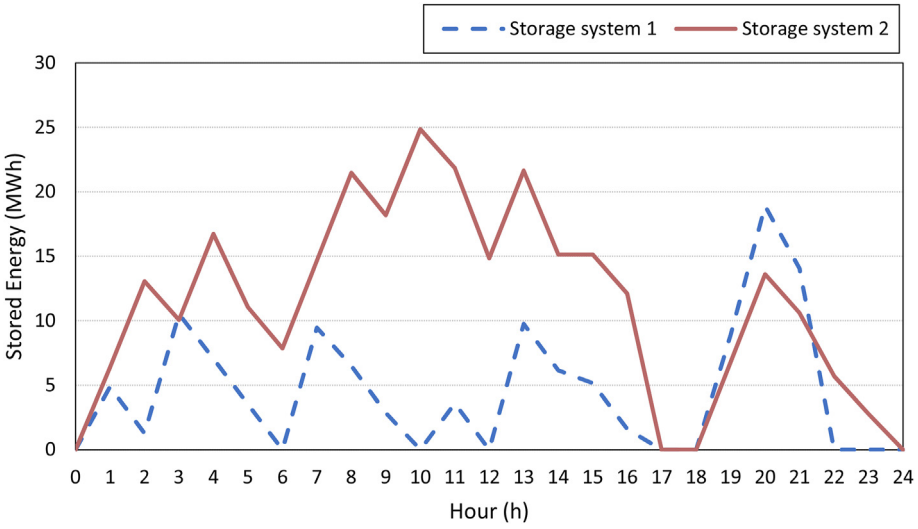


Fig. 5. State of charge of storage systems, for the uncongested network.

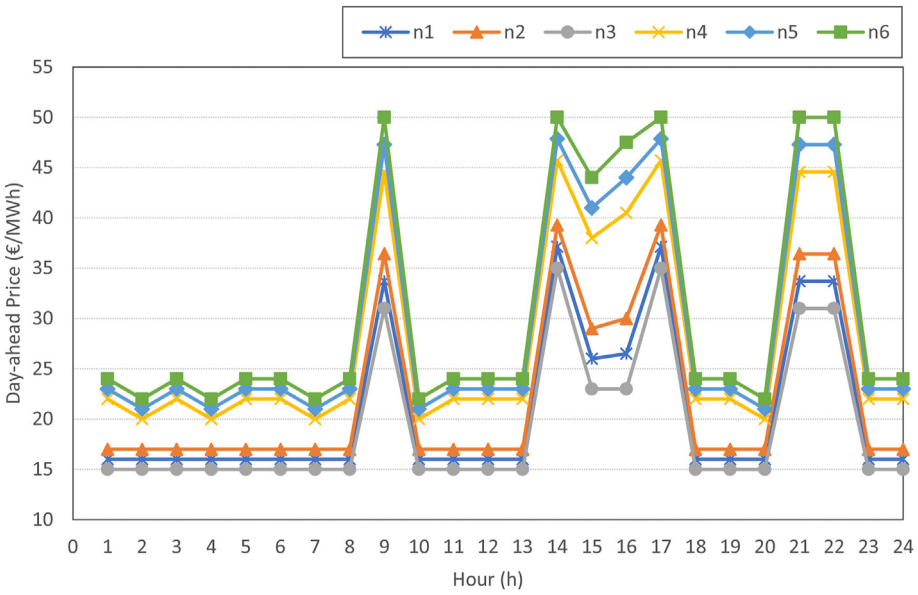


Fig. 6. Local marginal prices for the congested network.

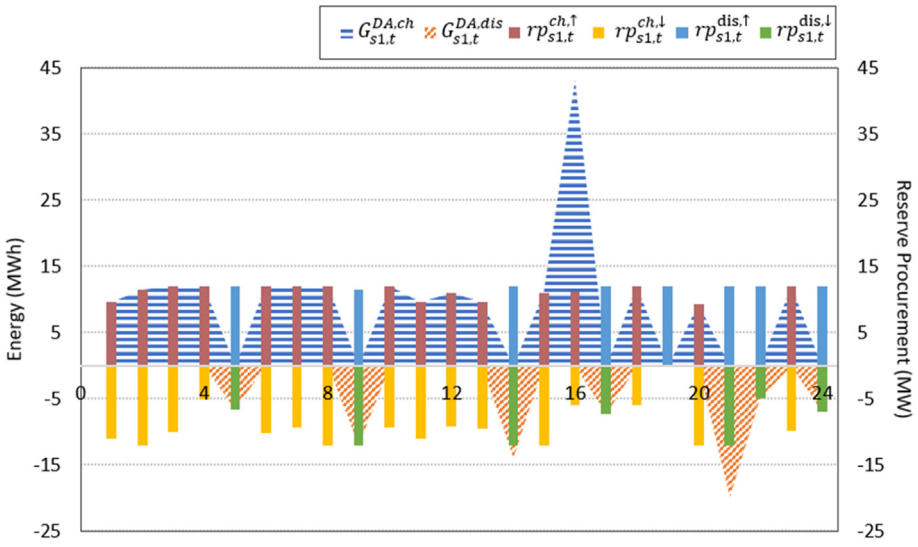


Fig. 7. Day-ahead (dis)charging energy dispatch and reserve procurements for storage system $s1$, for the congested network.

Due to the congested transmission line 3–6, energy transfer between these areas becomes extremely limited, thus leading to two discretized network settlements and resulting in different outcomes for the ESS agent, comparing to the uncongested network case. More specifically, storage system $s1$ located at bus $n1$ exploits this generation surplus occurred in the left area, by receiving more profitable (dis)charging decisions in order to maximize its revenues. These optimal strategies culminate in increased state-of-charge levels, as displayed in Fig. 8, in comparison to the equivalent quantities presented in Fig. 5. On the contrary, storage system $s2$ located in the low power generation subarea (bus $n6$), is qualified for managing restricted amount of energy comparing to the uncongested network case, which negatively affects its profit share, as it is analyzed below.

Table 3 to Table 4 and Table 5 to Table 6 summarize a comparison between the ESS agent's function in these two different networks. Table 3 presents the charging operation of storage system $s1$ for time period 7. More specifically, given the higher amounts of the energy charged and the increased day-ahead pool price, system $s1$ appears at first to experience profit losses under the congested network conditions. Analyzing the results further, the increase of upward reserve deployments at scenario $\omega 2$ (which are compensated at 200 €/MWh) with the parallel elimination of downward reserve deployments (which constitute a cost factor) seem to recast this conviction and crucially contribute to the $s1's$ revenue increment in the congested network. An equivalent attribute by $s1$ is

noticed in Table 4, regarding time period 14. Storage system $s1$ under congested power grid conditions, dispatches higher energy amounts in the day-ahead and real-time market, escalating the corresponding incomes and nullifies expenses originating from downward reserve activation in discharging mode, in comparison to the uncongested network. A reverse settlement is portrayed in Tables 5 and 6, regarding the strategies of storage system $s2$. System $s2$ charges less energy in the congested day-ahead market during time period 4, avoiding operational and purchasing expenses. Its insignificant participation in the real-time market though and particularly in activating upward reserves in the low wind generation scenario $\omega 2$, leads to profit depletion in comparison to $s2's$ profit generated in the uncongested network. Ultimately, during time period 16 and under congested network conditions, storage system $s2$ experiences zero revenues in the day-ahead and real-time energy market, in contradiction to the uncongested power grid case, where $s2$ increase its profit by activating upward reserves in discharging mode.

Fig. 9, through the triangular points on the green line, illustrates the positive or negative differentiations occurred in revenue for storage system $s1$, between the uncongested and congested network cases for an 24-h planning horizon and are measured in the right y-axis. On the other hand, the bars shown in the figure, correspond to energy amounts and are measured in MWh. Storage system $s1$ under congested network conditions, experiences losses due to an increase of day-ahead charging energy $G^{DA,ch}_{s1,t}$ and decrease of upward discharging reserve activation for low wind generation scenario $ra^{dis,\uparrow}_{s1,t,\omega 2}$, comparing to the uncongested network case. However, the quantitative increase of the remaining income increasing terms, such as $G^{DA,dis}_{s2,t1}$, $ra^{ch,\uparrow}_{s1,t,\omega 1}$, $ra^{ch,\uparrow}_{s1,t,\omega 2}$, contribute to a total profit increase by 6206 € in the congested network, as presented in Fig. 11. On the other hand, as demonstrated in Fig. 10, storage system $s2$, mainly due to the curtailed real-time upward charging and discharging reserve activation in scenario $\omega 2$ $(ra^{ch,\uparrow}_{s2,t,\omega 2}, ra^{dis,\uparrow}_{s2,t,\omega 2})$, generates significant profit losses up to 3406.6 €, comparing to the uncongested network settlement. In conclusion, despite the deficit of storage system $s2$, ESS agent, from the joint operation of storage systems s1 and s2, benefits from network congestions and experiences an overall profit increase by 2799.4 € compared to the uncongested case, as depicted in Fig. 11.

3.3. Wind power generation increment scenarios

Two additional cases of wind power generation increment are investigated, under uncongested network conditions, with Case 1 representing the base case with zero wind generation increase. Particularly, in *Case 2*, production of unit $j1$ rises from 70 MWh -which constitute the 12.17% of the grid's installed capacity-to 100 MWh in the high wind scenario $\omega 1$, thus increasing wind generation's share up to 16.53%, in the examined power system. As for the low wind scenario $\omega 2$, wind penetration increases to 20 MWh. These alterations result to less generation of expensive conventional units, as ISO favors the augmented energy provision by unit $j1$, due to its zero marginal cost offers. Specifically, unit's $i3$ and $i4$ dispatch in the day-ahead market is reduced by 56.2% and 15.1% respectively, due to their increased cost offers during the 24-h horizon, as shown in Table S1. Concerning *Case 3*, wind power provision reaches up to 19.2% of network's installed capacity, by dispatching 120 MWh and 20 MWh in the high and low wind scenario correlatively and de novo restricts conventional generators' participation in the energy day-ahead and real-time market.

Table S.16 depicts storage system's $s1$ response in the above analyzed wind increment cases, during time period 1. While additional energy originated from non-dispatchable producers accesses the power system, storage system $s1$, acting as a demand load, charges higher amounts of energy in the day-ahead and activates increased quantities of upward reserves in the real-time pool, as the market prices de-escalate, due to the zero marginal cost offers by $j1$. Characteristically, day-ahead price $\lambda^{DA}_{n,t1}$ is reduced from 15 €/MWh to 14 €/MWh, while the balancing price $\lambda^{RT}_{n,t1,\omega 1}$ for the high wind generation scenario $\omega 1$ decreases from 16 €/MWh to 14 €/MWh. Fig. 12 illustrates the differentiations in storage system's $s1$ energy dispatch and reserve procurements, occurred by the different wind generation increment cases. In *Case 2* and *Case 3*, a parallel increase of energy charged, and a decrease of energy discharged in the day-ahead market appear. At first, this realization could lead to the conclusion that $s1$ suffers significant profit losses while wind penetration intensifies, as $G^{DA,ch}_{s1,t}$ and $G^{DA,dis}_{s1,t}$ correspond

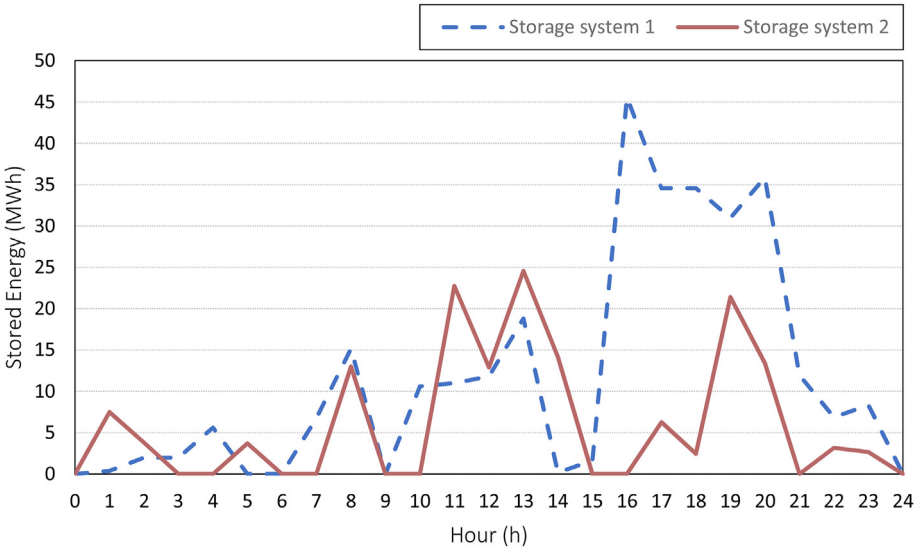


Fig. 8. State of charge of storage systems, for the congested network.

Table 3
Storage system's $s1$ operation in charging mode during time period 7.

	$G^{DA,ch}_{s1,t7}$ (MWh)	$rp^{ch,\downarrow}_{s1,t7}$ (MW)	$rp^{ch,\uparrow}_{s1,t7}$ (MW)	$ra^{ch,\downarrow}_{s1,t7,\omega1}$ (MWh)	$ra^{ch,\downarrow}_{s1,t7,\omega2}$ (MWh)	$ra^{ch,\uparrow}_{s1,t7,\omega1}$ (MWh)	$ra^{ch,\uparrow}_{s1,t7,\omega2}$ (MWh)	$\lambda^{DA}_{n1,t7}$ (€/MWh)	$\lambda^{RT}_{n1,t7,\omega1}$ (€/MWh)	$\lambda^{RT}_{n1,t1,\omega2}$ (€/MWh)
Uncongested network	4.3	12	3.83	12	–	–	3.83	15	16	200
Congested network	12	9.35	12	–	–	2.36	12	16	16	200

Table 4
Storage system's $s1$ operation in discharging mode during time period 14.

	$G^{DA,dis}_{s1,t14}$ (MWh)	$rp^{dis,\downarrow}_{s1,t14}$ (MW)	$rp^{dis,\uparrow}_{s1,t14}$ (MW)	$ra^{dis,\downarrow}_{s1,t14,\omega1}$ (MWh)	$ra^{dis,\downarrow}_{s1,t14,\omega2}$ (MWh)	$ra^{dis,\uparrow}_{s1,t14,\omega1}$ (MWh)	$ra^{dis,\uparrow}_{s1,t14,\omega2}$ (MWh)	$\lambda^{DA}_{n1,t14}$ (€/MWh)	$\lambda^{RT}_{n1,t14,\omega1}$ (€/MWh)	$\lambda^{RT}_{n1,t14,\omega2}$ (€/MWh)
Uncongested network	8.3	8.3	3.7	8.3	–	–	3.7	35	41	41
Congested network	14	12	12	–	–	1.6	12	37.14	41	41

Table 5
Storage system's $s2$ operation in charging mode during time period 4.

	$G^{DA,ch}_{s2,t4}$ (MWh)	$rp^{ch,\downarrow}_{s2,t4}$ (MW)	$rp^{ch,\uparrow}_{s2,t4}$ (MW)	$ra^{ch,\downarrow}_{s2,t4,\omega1}$ (MWh)	$ra^{ch,\downarrow}_{s2,t4,\omega2}$ (MWh)	$ra^{ch,\uparrow}_{s2,t4,\omega1}$ (MWh)	$ra^{ch,\uparrow}_{s2,t4,\omega2}$ (MWh)	$\lambda^{DA}_{n6,t4}$ (€/MWh)	$\lambda^{RT}_{n6,t4,\omega1}$ (€/MWh)	$\lambda^{RT}_{n6,t4,\omega2}$ (€/MWh)
Uncongested network	10	10	10	–	–	0.5	10	15	16	200
Congested network	–	7.8	–	–	–	–	–	22	16	200

Table 6
Storage system's $s2$ operation in discharging mode during time period 16.

	$G^{DA,dis}_{s2,t16}$ (MWh)	$rp^{dis,\downarrow}_{s2,t16}$ (MW)	$rp^{dis,\uparrow}_{s2,t16}$ (MW)	$ra^{dis,\downarrow}_{s2,t16,\omega1}$ (MWh)	$ra^{dis,\downarrow}_{s2,t16,\omega2}$ (MWh)	$ra^{dis,\uparrow}_{s2,t16,\omega1}$ (MWh)	$ra^{dis,\uparrow}_{s2,t16,\omega2}$ (MWh)	$\lambda^{DA}_{n6,t16}$ (€/MWh)	$\lambda^{RT}_{n6,t16,\omega1}$ (€/MWh)	$\lambda^{RT}_{n6,t16,\omega2}$ (€/MWh)
Uncongested network	–	0.03	10	–	–	–	10	24	26	26
Congested network	–	–	–	–	–	–	–	47.5	26	26

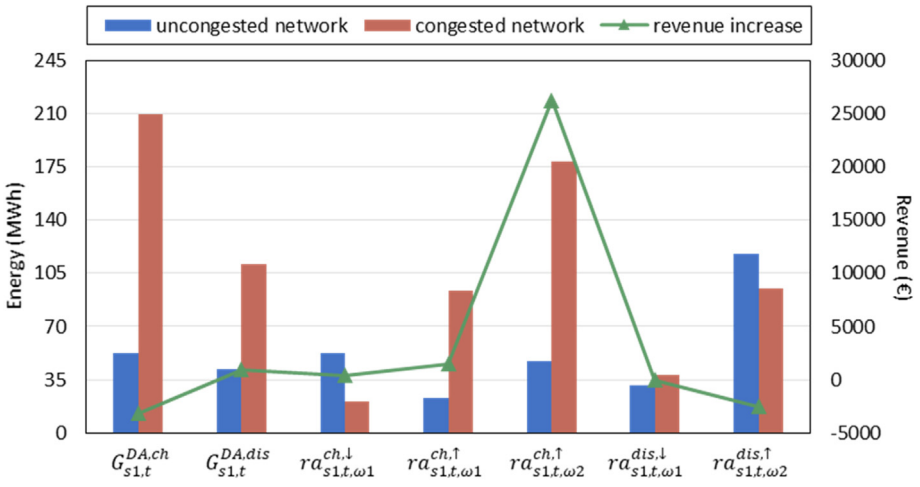


Fig. 9. Energy dispatch and revenue comparison for storage system $s1$, between uncongested and congested network.

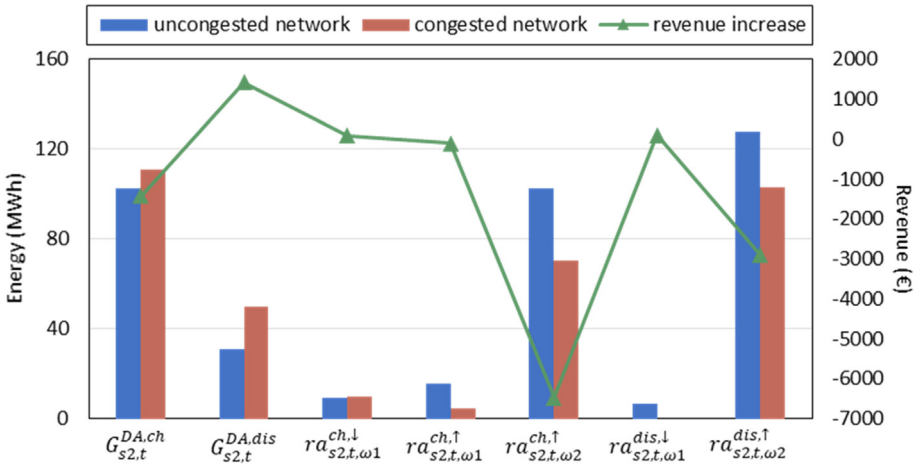


Fig. 10. Energy dispatch and revenue comparison for storage system *s*2, between uncongested and congested network.

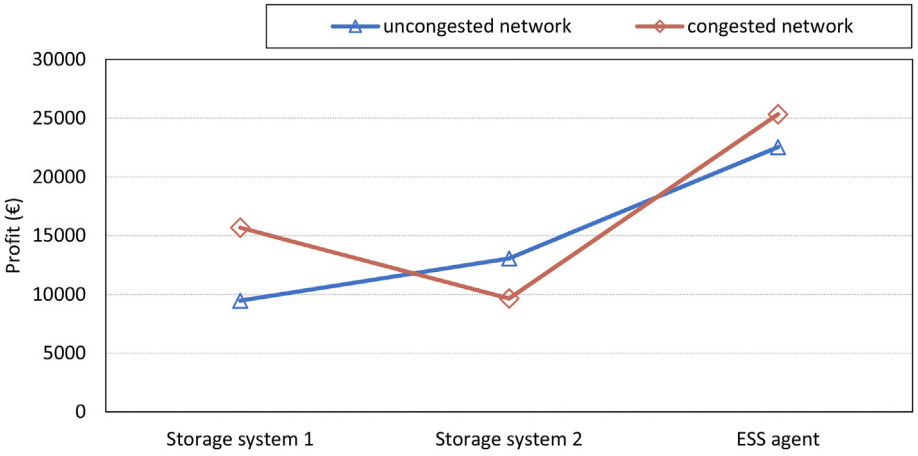


Fig. 11. Profit comparison between uncongested and congested network.

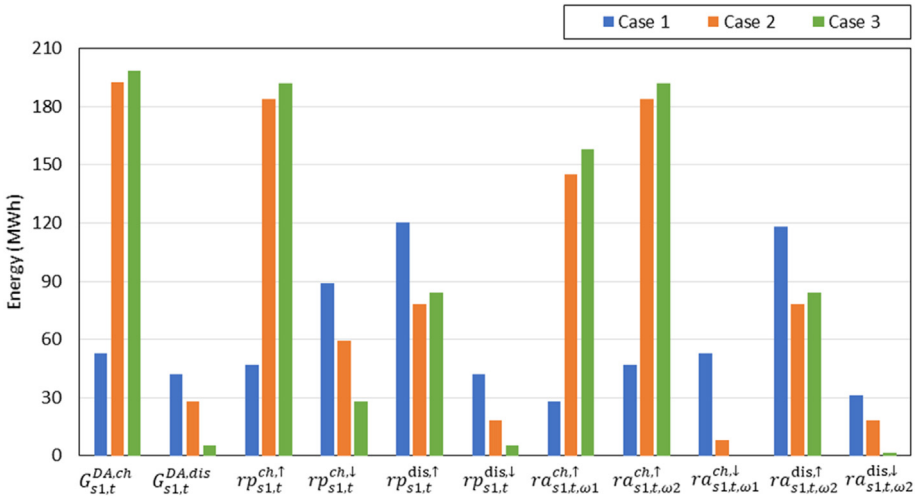


Fig. 12. Overall energy dispatch and reserve procurements of storage system *s*1, under three wind generation cases.

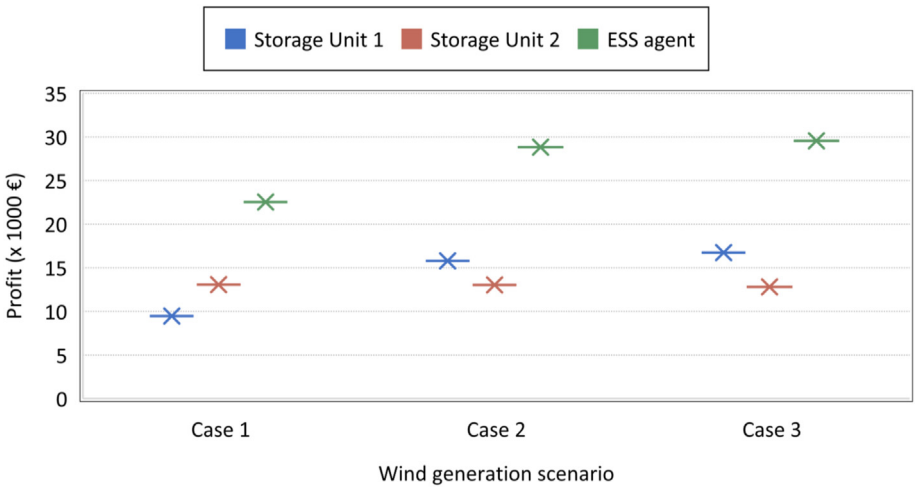


Fig. 13. Profit differentiation for storage systems $s1, s2$ and ESS agent, under three wind generation cases.

to an expense and revenue factor apiece. However, the assertive involvement of $s1$ in balancing market and especially by activating upward reserves in charging mode, compensated by market prices up to 200 €/MWh, leads to a critical source of income, which dominates any potential losses.

This claim is unambiguously justified in Fig. 13, as storage system's $s1$ profits systematically rise up to 15,792 € and 16,734 € in *Case 2* and *Case 3* equivalently. On the other hand, profit of storage system $s2$ is not considerably affected by the level of wind penetration. As for the ESS agent in total, mainly due to the unit's $s1$ optimal operation, experiences profit surplus by 6293 € in *Case 2* and 7004 € in *Case 3*, in comparison to *Case 1*.

4. Computational issues

Te resulting MILP has been solved using CPLEX/GAMS 32.1.0, on an Intel Core i7-9700 processor at 3.00 GHz, with 32 GB RAM. The average CPU time for solving the Uncongested network case is 1150 s, while for the Congested network case, 1640 s. Computational time increases in accordance with the network sophistication, examined wind power generation scenarios and transmission lines congestions. Binary expansion approach [32,46] provides an additional critically challenging factor, regarding the computational requirements of the model, as each bilinear term is substituted by a set of multiple parameters. Finally, as already mentioned the determination of big-Ms' value is firmly associated with the efficiency of the computation procedure.

5. Conclusions

This work, based on a single leader-follower Stackelberg hypothesis' game, illustrates a bi-level mathematical framework to generate optimal offering/bidding strategies for a strategic ESS agent in a pool-based market settlement. The model considers a sequential clearing mechanism of a joint energy and reserve day-ahead market and an energy-only real-time market, conducted by ISO. Uncertainty is introduced in the balancing stage through the realization of plausible wind power generation scenarios. The application of the proposed modelling approach on a 6-bus transmission constrained network, derives local marginal prices as dual variables of the lower-level problems and generates optimal dis (charging) energy dispatch and upward/downward reserve provisions for the ESS operator. The algorithm leads to an increase in clearing prices compared to cost optimization. Furthermore, the effects on ESS's functioning and expected profit inflow caused by potential network transmission congestions, are investigated. More specifically, the ESS agent benefits from line congestions and experiences significant profit increase, compared to the uncongested case. Conclusively, the model provides significant insights into designing policies that the energy storage agents should adopt, in order to capitalize a possible rise in the wind generation installed capacity and achieve an attractive profit increase.

A future research direction could focus on the development of an algorithm studying the strategic behavior of an agent with a mixed generation portfolio (i.e., conventional, wind and electricity storage) in a joint energy and reserve market. Furthermore, the market effects of a further 19.2% increase in wind energy production could be the subject of future research. Extensions could also integrate detailed technical parameters of storage and electricity generators or other types of renewables with a representative and realistic set of scenarios. In that case, due to the increased complexity of the optimization problem, special solution techniques (decomposition algorithms) should be investigated.

CRedit author statement

Christos N. Dimitriadis: Conceptualization, Investigation, Methodology, Software, Writing-Original draft. **Evangelos G. Tsimopoulos:** Conceptualization, Validation, Writing-Original draft. **Michael C. Georgiadis:** Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research has been co-financed by the European Union and Greek national funds through the Operational Program Competitiveness, Entrepreneurship and Innovation, under the call AQUACULTURES - INDUSTRIAL MATERIALS - OPEN INNOVATION IN CULTURE (project code: T6YBΠ-00251).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.energy.2021.123026.

References

[1] Green R, Vasilakos N. Market behaviour with large amounts of intermittent generation. Energy Pol 2010. https://doi.org/10.1016/j.enpol.2009.07.038.
[2] Denholm P, Ela E, Kirby B, Milligan M. The role of energy storage with renewable electricity generation. In: Energy storage: Issues and applications; 2011.
[3] International Renewable Energy Agency (IRENA). Electricity storage and renewables: costs and markets to 2030. 2017.
[4] Shafiee S, Zamani-Dehkordi P, Zareipour H, Knight AM. Economic assessment of a price-maker energy storage facility in the Alberta electricity market. Energy 2016. https://doi.org/10.1016/j.energy.2016.05.086.
[5] Sioshansi R, Denholm P, Jenkin T, Weiss J. Estimating the value of electricity storage in PJM: arbitrage and some welfare effects. Energy Econ 2009. https://doi.org/10.1016/j.eneco.2008.10.005.
[6] McConnell D, Forcey T, Sandiford M. Estimating the value of electricity storage in an energy-only wholesale market. Appl Energy 2015. https://doi.org/10.1016/j.apenergy.2015.09.006.
[7] Sioshansi FP. Competitive electricity markets: questions remain about design, implementation, performance. Electr J 2008. https://doi.org/10.1016/j.tej.2008.02.001.
[8] Dimitriadis CN, Tsimopoulos EG, Georgiadis MC. A review on the complementarity modelling in competitive electricity markets. Energies 2021;14(21). https://doi.org/10.3390/en14217133.
[9] Gabriel Steven A, et al. Complementarity modeling in energy markets. *Media*; 2013. https://doi.org/10.1007/978-1-4419-6123-5.
[10] Ruiz C, Conejo AJ. Pool strategy of a producer with endogenous formation of locational marginal prices. IEEE Trans Power Syst 2009. https://doi.org/10.1109/TPWRS.2009.2030378.
[11] Baringo L, Conejo AJ. Wind power investment within a market environment. Appl Energy 2011. https://doi.org/10.1016/j.apenergy.2011.03.023.
[12] Morales JM, Conejo AJ, Liu K, Zhong J. Pricing electricity in pools with wind producers. IEEE Trans Power Syst 2012. https://doi.org/10.1109/TPWRS.2011.2182622.
[13] Nasiri N, Zeynali S, Ravadanegh SN, Marzband M. A hybrid robust-stochastic approach for strategic scheduling of a multi-energy system as a price-maker player in day-ahead wholesale market. Energy 2021;235. https://doi.org/10.1016/j.energy.2021.121398.
[14] Tsimopoulos EG, Georgiadis MC. Strategic offers in day-ahead market co-optimizing energy and reserve under high penetration of wind power production: an MPEC approach. AIChE J 2019;65(7):1–11. https://doi.org/10.1002/aic.16495.
[15] Tsimopoulos EG, Georgiadis MC. Optimal strategic offerings for a conventional producer in jointly cleared energy and balancing markets under high penetration of wind power production. Appl Energy 2019. https://doi.org/10.1016/j.apenergy.2019.03.161.
[16] Awad ASA, Fuller JD, El-Fouly THM, Salama MMA. Impact of energy storage systems on electricity market equilibrium. IEEE Trans Sustain Energy 2014. https://doi.org/10.1109/TSTE.2014.2309661.
[17] Pandžić H, Kuzle I. Energy storage operation in the day-ahead electricity market. 2015. https://doi.org/10.1109/EEM.2015.7216754.
[18] Ye Y, Papadaskalopoulos D, Strbac G. An MPEC approach for analysing the impact of energy storage in imperfect electricity markets. 2016. https://doi.org/10.1109/EEM.2016.7521287.
[19] Wang Y, Dvorkin Y, Fernández-Blanco R, Xu B, Qiu T, Kirschen DS. Look-ahead bidding strategy for energy storage. IEEE Trans Sustain Energy 2017. https://doi.org/10.1109/TSTE.2017.2656800.
[20] Gomes ILR, Pousinho HMI, Melício R, Mendes VMF. Stochastic coordination of joint wind and photovoltaic systems with energy storage in day-ahead market. Energy 2017;124:310–20. https://doi.org/10.1016/j.energy.2017.02.080.
[21] Ding H, Pinson P, Hu Z, Wang J, Song Y. Optimal offering and operating strategy for a large wind-storage system as a price maker. IEEE Trans Power Syst 2017. https://doi.org/10.1109/TPWRS.2017.2681720.
[22] Zou P, Chen Q, Xia Q, He G, Kang C, Conejo AJ. Pool equilibria including strategic storage. Appl Energy 2016. https://doi.org/10.1016/j.apenergy.2016.05.105.
[23] Shahmohammadi A, Sioshansi R, Conejo AJ, Afsharnia S. Market equilibria and interactions between strategic generation, wind, and storage. Appl Energy 2018. https://doi.org/10.1016/j.apenergy.2017.10.035.
[24] Wang J, Xin H, Xie N, Wang Y. Equilibrium models of coordinated electricity and natural gas markets with different coupling information exchanging channels. Energy 2022;239:121827. https://doi.org/10.1016/j.energy.2021.121827.
[25] Tsimopoulos EG, Georgiadis MC. Nash equilibria in electricity pool markets with large-scale wind power integration. Energy 2021;228:120642. https://doi.org/10.1016/j.energy.2021.120642.
[26] González P, Villar J, Díaz CA, Campos FA. Joint energy and reserve markets: current implementations and modeling trends. Elec Power Syst Res 2014. https://doi.org/10.1016/j.epsr.2013.12.013.
[27] Drury E, Denholm P, Sioshansi R. The value of compressed air energy storage in energy and reserve markets. Energy 2011. https://doi.org/10.1016/j.energy.2011.05.041.
[28] Reddy SS, Bijwe PR, Abhyankar AR. Optimum day-ahead clearing of energy and reserve markets with wind power generation using anticipated real-time adjustment costs. Int J Electr Power Energy Syst 2015. https://doi.org/10.1016/j.ijepes.2015.03.002.
[29] Nojavan S, Najafi-Ghalelou A, Majidi M, Zare K. Optimal bidding and offering strategies of merchant compressed air energy storage in deregulated electricity market using robust optimization approach. Energy 2018;142:250–7. https://doi.org/10.1016/j.energy.2017.10.028.
[30] Tsimopoulos EG, Georgiadis MC. Withholding strategies for a conventional and wind generation portfolio in a joint energy and reserve pool market: a gaming-based approach. Comput Chem Eng 2020. https://doi.org/10.1016/j.compchemeng.2019.106692.
[31] Vespermann N, Delikaraoglou S, Pinson P. Offering strategy of a price-maker energy storage system in day-ahead and balancing markets. 2017. https://doi.org/10.1109/PTC.2017.7981050.
[32] Nasrolahpour E, Kazempour J, Zareipour H, Rosehart WD. A bilevel model for participation of a storage system in energy and reserve markets. IEEE Trans Sustain Energy 2018. https://doi.org/10.1109/TSTE.2017.2749434.
[33] Ruiz C, Conejo AJ, Gabriel SA. Pricing non-convexities in an electricity pool. IEEE Trans Power Syst 2012. https://doi.org/10.1109/TPWRS.2012.2184562.
[34] Schillemans A, De Vivero Serrano G, De Vivero Serrano G, Bruninx K, Bruninx K, Bruninx K. Strategic participation of merchant energy storage in joint energy-reserve and balancing markets. 2018. https://doi.org/10.1049/cp.2018.1920.
[35] Khojasteh M, Faria P, Vale Z. A robust model for aggregated bidding of energy storages and wind resources in the joint energy and reserve markets. Energy 2022;238:121735. https://doi.org/10.1016/j.energy.2021.121735.
[36] Khaloie H, et al. Coordinated wind-thermal-energy storage offering strategy in energy and spinning reserve markets using a multi-stage model. Appl Energy 2020. https://doi.org/10.1016/j.apenergy.2019.114168.
[37] Mohsenian-Rad H. Coordinated price-maker operation of large energy storage units in nodal energy markets. IEEE Trans Power Syst 2016;31(1):786–97. https://doi.org/10.1109/TPWRS.2015.2411556.
[38] California Public Utility Commission. Energy storage decision D.13-10-040. *Oct.2013 [Online]. Available:*, https://www.cpuc.ca.gov/.
[39] Vega-Redondo F. Economics and the theory of games. 2003.
[40] Pennsylvania-New Jersey-Maryland Interconnection (PJM). PJM manual 11: energy & ancillary services market operations. *Mar. 2016. [Online]. Available:*, https://www.pjm.com/~/media/documents/manuals/m11.ashx.
[41] Pennsylvania-New Jersey-Maryland Interconnection (PJM). Renewable integration study. *Apr. 2013. [Online]. Available:*, https://www.pjm.com/~/media/committees-groups/subcommittees/irs/postings/pris-task3b-best-practices-from-other-markets-final-report.ashx.
[42] Morales JM, Conejo AJ, Madsen H, Pinson P, Zugno M. Integrating renewables in electricity markets - operational problems. 2014.
[43] S. Stoft, Power system economics: designing markets for electricity, vol. 99, no. 1. 2003.
[44] Conejo AJ, Carrión M, Morales JM. Decision making under Uncertainty in electricity markets. 2010. 978-1-4419-7421–1 Springer.
[45] Fortuny-Amat J, McCarl B. A representation and economic interpretation of a two-level programming problem. J Oper Res Soc 1981. https://doi.org/10.2307/2581394.
[46] Pereira MV, Granville S, Fampa MHC, Dix R, Barroso LA. Strategic bidding under uncertainty: a binary expansion approach. IEEE Trans Power Syst 2005. https://doi.org/10.1109/TPWRS.2004.840397.
[47] Brook A, Kendrick D, Meeraus A. GAMS, a user's guide. ACM SIGNUM Newsl. 1988. https://doi.org/10.1145/58859.58863.